tropical: MinCoinChange

2026-10-25T15:59:14

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `7237e0b98779`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.25	[0.232, 0.267]	0.0208	5	0
adaptive	pointer	0.01	0.2547	[0.244, 0.268]	0.0149	5	0
adaptive	pointer	0.1	0.2547	[0.245, 0.267]	0.0151	5	0
adaptive	pointer	1	0.2557	[0.245, 0.27]	0.0152	5	0
adaptive	pointer	10	0.2559	[0.246, 0.268]	0.0148	5	0
adaptive	pointer	100	0.2555	[0.247, 0.268]	0.0145	5	0
adaptive	pointer	1000	0.2547	[0.246, 0.267]	0.0143	5	0
adaptive	pointer	1e+04	0.2545	[0.245, 0.267]	0.0137	5	0
adaptive	seq2seq	0	0.2208	[0.212, 0.23]	0.0119	5	0
adaptive	seq2seq	0.01	0.2212	[0.21, 0.232]	0.0146	5	0
adaptive	seq2seq	0.1	0.2152	[0.195, 0.233]	0.0238	5	0
adaptive	seq2seq	1	0.235	[0.223, 0.248]	0.0162	5	0
adaptive	seq2seq	10	0.2602	[0.225, 0.291]	0.0444	5	0
adaptive	seq2seq	100	0.1943	[0.177, 0.214]	0.0244	5	0
adaptive	seq2seq	1000	0.1843	[0.0935, 0.236]	0.103	5	0
adaptive	seq2seq	1e+04	0.1843	[0.0904, 0.236]	0.103	5	0
tropical	pointer	0	0.2757	[0.258, 0.295]	0.0239	5	0
tropical	pointer	0.01	0.2754	[0.254, 0.297]	0.0279	5	0
tropical	pointer	0.1	0.2711	[0.256, 0.287]	0.0201	5	0
tropical	pointer	1	0.2739	[0.259, 0.289]	0.02	5	0
tropical	pointer	10	0.2741	[0.259, 0.291]	0.0214	5	0
tropical	pointer	100	0.2732	[0.258, 0.288]	0.0191	5	0
tropical	pointer	1000	0.2822	[0.259, 0.305]	0.0298	5	0
tropical	pointer	1e+04	0.2741	[0.262, 0.286]	0.0161	5	0
tropical	seq2seq	0	0.2017	[0.174, 0.23]	0.0375	5	0
tropical	seq2seq	0.01	0.2186	[0.186, 0.252]	0.0419	5	0
tropical	seq2seq	0.1	0.2227	[0.204, 0.241]	0.0236	5	0
tropical	seq2seq	1	0.2287	[0.208, 0.25]	0.0288	5	0
tropical	seq2seq	10	0.2463	[0.212, 0.281]	0.044	5	0
tropical	seq2seq	100	0.2274	[0.211, 0.243]	0.0216	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.2212	[0.2, 0.238]	0.0254	5	0
tropical	seq2seq	1e+04	0.2212	[0.201, 0.238]	0.0254	5	0
vanilla	pointer	0	0.2399	[0.211, 0.271]	0.039	5	0
vanilla	pointer	0.01	0.2415	[0.211, 0.272]	0.0395	5	0
vanilla	pointer	0.1	0.242	[0.21, 0.272]	0.04	5	0
vanilla	pointer	1	0.243	[0.213, 0.271]	0.0384	5	0
vanilla	pointer	10	0.243	[0.208, 0.272]	0.0389	5	0
vanilla	pointer	100	0.2424	[0.212, 0.271]	0.0394	5	0
vanilla	pointer	1000	0.2426	[0.213, 0.271]	0.0378	5	0
vanilla	pointer	1e+04	0.2437	[0.216, 0.271]	0.0366	5	0
vanilla	seq2seq	0	0.2137	[0.207, 0.221]	0.00898	5	0
vanilla	seq2seq	0.01	0.2142	[0.209, 0.22]	0.00708	5	0
vanilla	seq2seq	0.1	0.2159	[0.204, 0.226]	0.0137	5	0
vanilla	seq2seq	1	0.2184	[0.21, 0.227]	0.0114	5	0
vanilla	seq2seq	10	0.2366	[0.214, 0.26]	0.0289	5	0
vanilla	seq2seq	100	0.2034	[0.18, 0.237]	0.0366	5	0
vanilla	seq2seq	1000	0.2364	[0.226, 0.246]	0.0133	5	0
vanilla	seq2seq	1e+04	0.2366	[0.226, 0.246]	0.0132	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.004722	[-0.0224, 0.0319]	0.9964
adaptive	pointer	0.1	0.004689	[-0.0225, 0.0319]	0.9965
adaptive	pointer	1	0.005729	[-0.0214, 0.0329]	0.9889
adaptive	pointer	10	0.005918	[-0.0212, 0.0331]	0.9867
adaptive	pointer	100	0.005523	[-0.0216, 0.0327]	0.9909
adaptive	pointer	1000	0.004699	[-0.0225, 0.0319]	0.9965
adaptive	pointer	1e+04	0.004518	[-0.0226, 0.0317]	0.9972
adaptive	seq2seq	0.01	4.315×10^{-4}	[-0.0975, 0.0984]	1.0000
adaptive	seq2seq	0.1	-0.005603	[-0.104, 0.0923]	1.0000
adaptive	seq2seq	1	0.01428	[-0.0837, 0.112]	0.9988
adaptive	seq2seq	10	0.03939	[-0.0585, 0.137]	0.7880
adaptive	seq2seq	100	-0.02646	[-0.124, 0.0715]	0.9592
adaptive	seq2seq	1000	-0.03646	[-0.134, 0.0615]	0.8385
adaptive	seq2seq	1e+04	-0.03646	[-0.134, 0.0615]	0.8386
tropical	pointer	0.01	-3.323×10^{-4}	[-0.04, 0.0394]	1.0000
tropical	pointer	0.1	-0.004586	[-0.0443, 0.0351]	0.9997
tropical	pointer	1	-0.001881	[-0.0416, 0.0378]	1.0000
tropical	pointer	10	-0.001621	[-0.0413, 0.0381]	1.0000
tropical	pointer	100	-0.002576	[-0.0423, 0.0371]	1.0000
tropical	pointer	1000	0.006478	[-0.0332, 0.0462]	0.9975
tropical	pointer	1e+04	-0.001668	[-0.0414, 0.038]	1.0000
tropical	seq2seq	0.01	0.01693	[-0.0391, 0.073]	0.9313
tropical	seq2seq	0.1	0.021	[-0.0351, 0.0771]	0.8350
tropical	seq2seq	1	0.02696	[-0.0291, 0.083]	0.6390
tropical	seq2seq	10	0.04463	[-0.0114, 0.101]	0.1644
tropical	seq2seq	100	0.02569	[-0.0304, 0.0818]	0.6833
tropical	seq2seq	1000	0.01954	[-0.0365, 0.0756]	0.8746
tropical	seq2seq	1e+04	0.01954	[-0.0365, 0.0756]	0.8746
vanilla	pointer	0.01	0.001619	[-0.066, 0.0693]	1.0000
vanilla	pointer	0.1	0.002041	[-0.0656, 0.0697]	1.0000
vanilla	pointer	1	0.003061	[-0.0646, 0.0707]	1.0000

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	10	0.003066	[-0.0646, 0.0707]	1.0000
vanilla	pointer	100	0.002457	[-0.0652, 0.0701]	1.0000
vanilla	pointer	1000	0.00273	[-0.0649, 0.0704]	1.0000
vanilla	pointer	1e+04	0.003753	[-0.0639, 0.0714]	1.0000
vanilla	seq2seq	0.01	5.051×10^{-4}	[-0.0332, 0.0342]	1.0000
vanilla	seq2seq	0.1	0.002118	[-0.0316, 0.0358]	1.0000
vanilla	seq2seq	1	0.004662	[-0.029, 0.0384]	0.9991
vanilla	seq2seq	10	0.02287	[-0.0108, 0.0566]	0.2952
vanilla	seq2seq	100	-0.01038	[-0.0441, 0.0233]	0.9252
vanilla	seq2seq	1000	0.02265	[-0.0111, 0.0564]	0.3041
vanilla	seq2seq	1e+04	0.02287	[-0.0108, 0.0566]	0.2948

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-2.855×10^{-5}	$[-3.83 \times 10^{-4}, 2.89 \times 10^{-4}]$	0.9802	-0.0552	0.6755
adaptive	seq2seq	-0.007612	[-0.0244, 0.00398]	0.1379	-0.0305	0.8214
tropical	pointer	6.223×10^{-4}	$[-3.40 \times 10^{-5}, 0.00128]$	0.7281	0.101	0.4661
tropical	seq2seq	1.287×10^{-4}	[-0.00533, 0.00559]	0.9600	0.0114	0.9380
vanilla	pointer	2.562×10^{-4}	$[-1.35 \times 10^{-4}, 6.13 \times 10^{-4}]$	0.9327	0.0324	0.8278
vanilla	seq2seq	0.003326	$[9.08 \times 10^{-4}, 0.006]$	0.0781	0.246	0.0602

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.03582	[-0.0177, 0.0894]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.03387	[-0.0354, 0.103]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.0292	[-0.0331, 0.0915]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.03088	[-0.0271, 0.0889]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.03113	[-0.0352, 0.0975]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.03079	[-0.0298, 0.0914]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.03957	[-0.0277, 0.107]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0304	[-0.0267, 0.0875]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.01203	[-0.0662, 0.0421]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.004399	[-0.0507, 0.0595]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.006853	[-0.0306, 0.0444]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01027	[-0.0146, 0.0352]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.009735	[-0.0413, 0.0608]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.02403	[-0.0445, 0.0926]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.01514	[-0.0484, 0.0182]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.01537	[-0.0486, 0.0178]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01009	[-0.0227, 0.0429]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01319	[-0.0367, 0.0631]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.01274	[-0.0377, 0.0632]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01276	[-0.0356, 0.0611]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01294	[-0.0373, 0.0632]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.01315	[-0.0357, 0.062]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01206	[-0.034, 0.0581]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01085	[-0.0351, 0.0568]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.007029	[-0.00837, 0.0224]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.006955	[-0.00667, 0.0206]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-6.922×10^{-4}	[-0.0338, 0.0324]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.01664	[-0.0047, 0.038]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.02355	[-0.0389, 0.086]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.009058	[-0.0748, 0.0567]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.05208	[-0.178, 0.0743]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.05231	[-0.179, 0.0747]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.02923	[0.00157, 0.0569]	0.7348
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.03352	[0.00705, 0.06]	0.5156
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03953	[0.00895, 0.0701]	0.5056
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.02069	[-0.0118, 0.0532]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.00424	[-0.0603, 0.0518]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.06122	[0.0305, 0.092]	0.1258
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.07039	[-0.051, 0.192]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.07021	[-0.0488, 0.189]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.07403	[0.00205, 0.146]	0.7348
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.05676	[0.00206, 0.111]	0.7348
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.04844	[0.0147, 0.0822]	0.3765
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.04519	[0.00787, 0.0825]	0.5367
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.02777	[-0.0378, 0.0934]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.04576	[0.00868, 0.0828]	0.5324
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.06097	$[-9.64 \times 10^{-4}, 0.123]$	0.7348
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.05282	[0.00357, 0.102]	0.7348
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02617	[-0.0319, 0.0843]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.02729	[-0.0257, 0.0803]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0261	[-0.0267, 0.0789]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.02457	[-0.0245, 0.0736]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.006374	[-0.074, 0.0867]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.03901	[-0.00397, 0.082]	0.8496
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.006255	[-0.0446, 0.0571]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.007053	[-0.0423, 0.0564]	1.0000

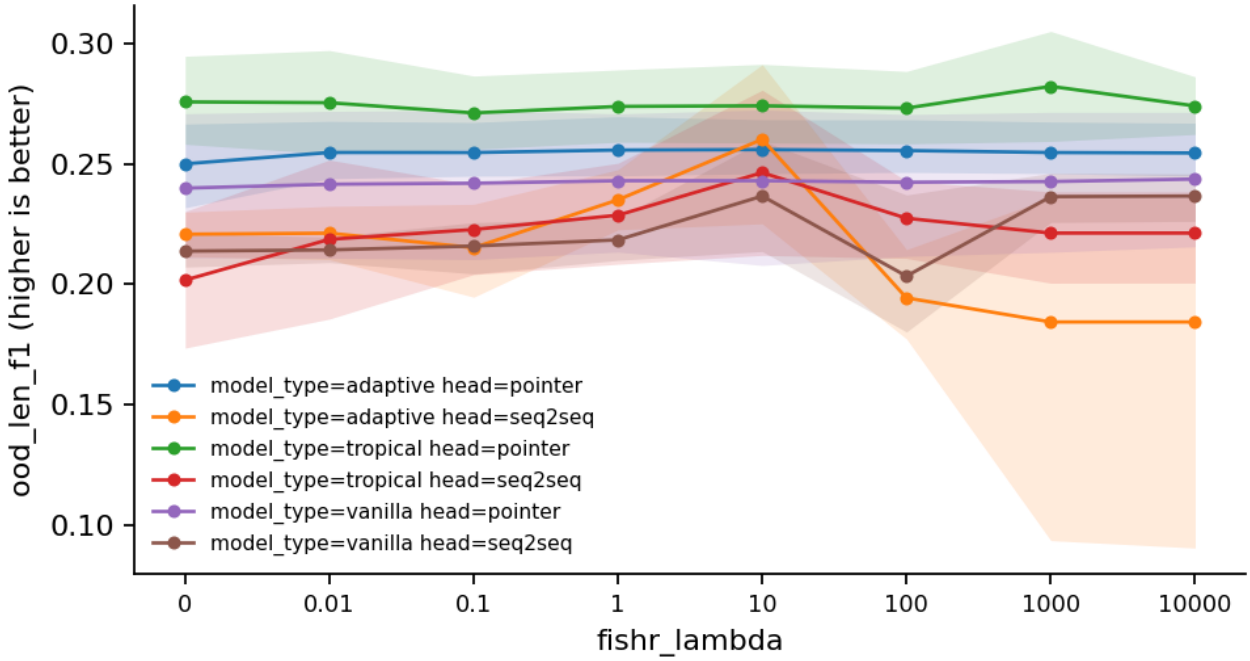


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

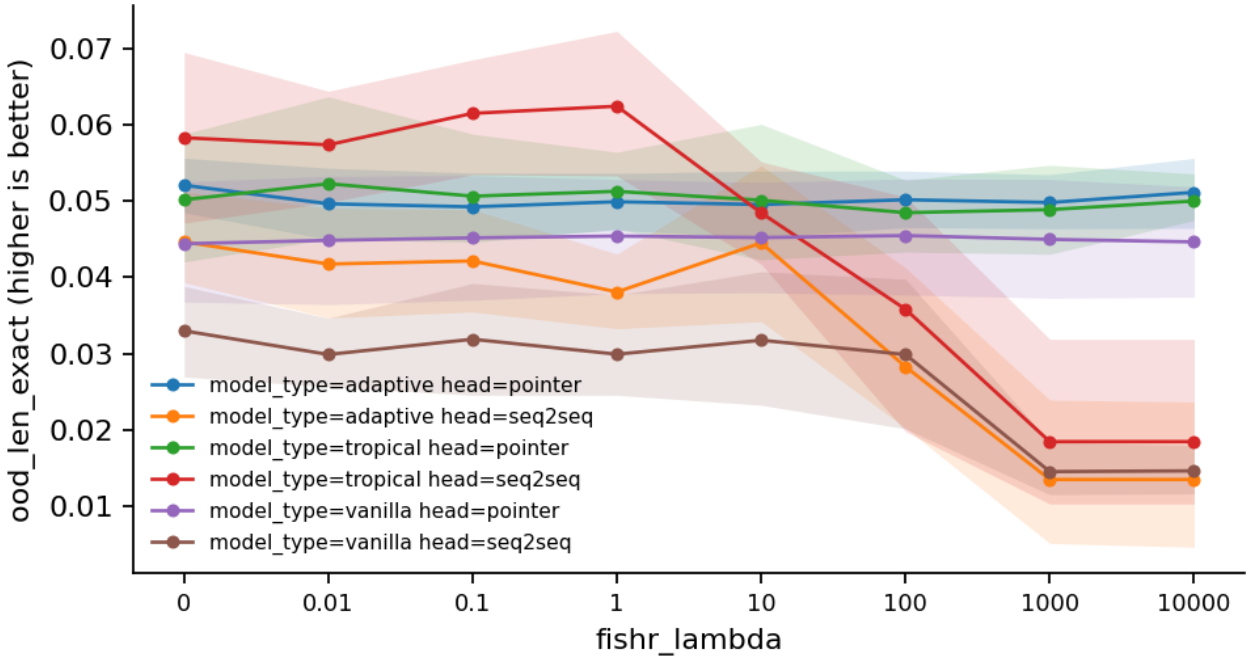


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.05207	[0.0485, 0.0556]	0.00491	5	0
adaptive	pointer	0.01	0.04963	[0.045, 0.0543]	0.00595	5	0
adaptive	pointer	0.1	0.04923	[0.045, 0.0537]	0.00555	5	0
adaptive	pointer	1	0.0499	[0.0462, 0.0536]	0.00485	5	0
adaptive	pointer	10	0.04953	[0.0453, 0.0539]	0.00569	5	0
adaptive	pointer	100	0.05017	[0.0465, 0.0539]	0.00485	5	0
adaptive	pointer	1000	0.0498	[0.0464, 0.0535]	0.00455	5	0
adaptive	pointer	1e+04	0.05113	[0.0464, 0.0556]	0.00608	5	0
adaptive	seq2seq	0	0.04463	[0.0393, 0.0512]	0.00756	5	0
adaptive	seq2seq	0.01	0.04173	[0.0347, 0.0491]	0.00973	5	0
adaptive	seq2seq	0.1	0.04213	[0.0355, 0.0489]	0.0088	5	0
adaptive	seq2seq	1	0.03807	[0.0333, 0.043]	0.00674	5	0
adaptive	seq2seq	10	0.0445	[0.0342, 0.0546]	0.0133	5	0
adaptive	seq2seq	100	0.02827	[0.0202, 0.0412]	0.0145	5	0
adaptive	seq2seq	1000	0.01347	[0.0051, 0.0239]	0.0124	5	0
adaptive	seq2seq	1e+04	0.01347	[0.0046, 0.0236]	0.0124	5	0
tropical	pointer	0	0.05017	[0.042, 0.0588]	0.0112	5	0
tropical	pointer	0.01	0.05227	[0.045, 0.0637]	0.0127	5	0
tropical	pointer	0.1	0.05063	[0.0446, 0.0588]	0.00968	5	0
tropical	pointer	1	0.05127	[0.0463, 0.0564]	0.00655	5	0
tropical	pointer	10	0.0501	[0.0423, 0.0601]	0.0114	5	0
tropical	pointer	100	0.04847	[0.0433, 0.0527]	0.00627	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.04887	[0.043, 0.0547]	0.00754	5	0
tropical	pointer	1e+04	0.05	[0.0475, 0.0536]	0.00394	5	0
tropical	seq2seq	0	0.0583	[0.0471, 0.0695]	0.0147	5	0
tropical	seq2seq	0.01	0.05737	[0.0498, 0.0644]	0.00993	5	0
tropical	seq2seq	0.1	0.0615	[0.0535, 0.0685]	0.00987	5	0
tropical	seq2seq	1	0.06243	[0.0533, 0.0723]	0.0125	5	0
tropical	seq2seq	10	0.04843	[0.0417, 0.0551]	0.00906	5	0
tropical	seq2seq	100	0.0358	[0.0198, 0.0505]	0.0202	5	0
tropical	seq2seq	1000	0.01843	[0.0103, 0.0319]	0.015	5	0
tropical	seq2seq	1e+04	0.01843	[0.0103, 0.0319]	0.015	5	0
vanilla	pointer	0	0.0444	[0.0368, 0.0525]	0.0103	5	0
vanilla	pointer	0.01	0.04483	[0.0364, 0.0532]	0.0108	5	0
vanilla	pointer	0.1	0.04517	[0.037, 0.0533]	0.0105	5	0
vanilla	pointer	1	0.0454	[0.0379, 0.0528]	0.00964	5	0
vanilla	pointer	10	0.0452	[0.038, 0.0524]	0.00924	5	0
vanilla	pointer	100	0.04547	[0.0377, 0.0529]	0.01	5	0
vanilla	pointer	1000	0.04497	[0.0373, 0.0528]	0.0101	5	0
vanilla	pointer	1e+04	0.04463	[0.0374, 0.0519]	0.00921	5	0
vanilla	seq2seq	0	0.03297	[0.0269, 0.0388]	0.00779	5	0
vanilla	seq2seq	0.01	0.02987	[0.0256, 0.0346]	0.00571	5	0
vanilla	seq2seq	0.1	0.03187	[0.0245, 0.0392]	0.00981	5	0
vanilla	seq2seq	1	0.0299	[0.0245, 0.0377]	0.00879	5	0
vanilla	seq2seq	10	0.03173	[0.0232, 0.0407]	0.0119	5	0
vanilla	seq2seq	100	0.02987	[0.0201, 0.0397]	0.0127	5	0
vanilla	seq2seq	1000	0.0145	[0.0115, 0.0184]	0.00429	5	0
vanilla	seq2seq	1e+04	0.0146	[0.0116, 0.0179]	0.00424	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.002433	[-0.0117, 0.00688]	0.9654	
adaptive	pointer	0.1	-0.002833	[-0.0121, 0.00648]	0.9291	
adaptive	pointer	1	-0.002167	[-0.0115, 0.00715]	0.9810	
adaptive	pointer	10	-0.002533	[-0.0118, 0.00678]	0.9579	
adaptive	pointer	100	-0.0019	[-0.0112, 0.00741]	0.9907	
adaptive	pointer	1000	-0.002267	[-0.0116, 0.00705]	0.9759	
adaptive	pointer	1e+04	-9.333×10^{-4}	[-0.0102, 0.00838]	0.9999	
adaptive	seq2seq	0.01	-0.0029	[-0.0221, 0.0163]	0.9985	
adaptive	seq2seq	0.1	-0.0025	[-0.0217, 0.0167]	0.9994	
adaptive	seq2seq	1	-0.006567	[-0.0258, 0.0127]	0.8842	
adaptive	seq2seq	10	-1.333×10^{-4}	[-0.0194, 0.0191]	1.0000	
adaptive	seq2seq	100	-0.01637	[-0.0356, 0.00286]	0.1216	
adaptive	seq2seq	1000	-0.03117	[-0.0504, -0.0119]	0.0006	*
adaptive	seq2seq	1e+04	-0.03117	[-0.0504, -0.0119]	0.0006	*
tropical	pointer	0.01	0.0021	[-0.0138, 0.018]	0.9993	
tropical	pointer	0.1	4.667×10^{-4}	[-0.0154, 0.0164]	1.0000	
tropical	pointer	1	0.0011	[-0.0148, 0.017]	1.0000	
tropical	pointer	10	-6.667×10^{-5}	[-0.016, 0.0158]	1.0000	
tropical	pointer	100	-0.0017	[-0.0176, 0.0142]	0.9998	
tropical	pointer	1000	-0.0013	[-0.0172, 0.0146]	1.0000	
tropical	pointer	1e+04	-1.667×10^{-4}	[-0.0161, 0.0157]	1.0000	
tropical	seq2seq	0.01	-9.333×10^{-4}	[-0.0249, 0.023]	1.0000	
tropical	seq2seq	0.1	0.0032	[-0.0208, 0.0272]	0.9993	

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	seq2seq	1	0.004133	[-0.0198, 0.0281]	0.9965	
tropical	seq2seq	10	-0.009867	[-0.0338, 0.0141]	0.7714	
tropical	seq2seq	100	-0.0225	[-0.0465, 0.00148]	0.0731	
tropical	seq2seq	1000	-0.03987	[-0.0638, -0.0159]	0.0004	*
tropical	seq2seq	1e+04	-0.03987	[-0.0638, -0.0159]	0.0004	*
vanilla	pointer	0.01	4.333×10^{-4}	[-0.017, 0.0179]	1.0000	
vanilla	pointer	0.1	7.667×10^{-4}	[-0.0167, 0.0182]	1.0000	
vanilla	pointer	1	1.000×10^{-3}	[-0.0165, 0.0185]	1.0000	
vanilla	pointer	10	8.000×10^{-4}	[-0.0167, 0.0183]	1.0000	
vanilla	pointer	100	0.001067	[-0.0164, 0.0185]	1.0000	
vanilla	pointer	1000	5.667×10^{-4}	[-0.0169, 0.018]	1.0000	
vanilla	pointer	1e+04	2.333×10^{-4}	[-0.0172, 0.0177]	1.0000	
vanilla	seq2seq	0.01	-0.0031	[-0.0183, 0.0121]	0.9908	
vanilla	seq2seq	0.1	-0.0011	[-0.0163, 0.0141]	1.0000	
vanilla	seq2seq	1	-0.003067	[-0.0183, 0.0121]	0.9913	
vanilla	seq2seq	10	-0.001233	[-0.0164, 0.014]	1.0000	
vanilla	seq2seq	100	-0.0031	[-0.0183, 0.0121]	0.9908	
vanilla	seq2seq	1000	-0.01847	[-0.0337, -0.00326]	0.0120	*
vanilla	seq2seq	1e+04	-0.01837	[-0.0336, -0.00316]	0.0126	*

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.107×10^{-4}	$[3.93 \times 10^{-5}, 3.69 \times 10^{-4}]$	0.6199	0.0743	0.5783
adaptive	seq2seq	-0.005426	[-0.00672, -0.00413]	$< 10^{-4}$	-0.533	0.0002
tropical	pointer	-4.690×10^{-4}	$[-0.00163, 1.75 \times 10^{-4}]$	0.4988	0.0133	0.9296
tropical	seq2seq	-0.008199	[-0.00929, -0.00696]	$< 10^{-4}$	-0.617	0.0002
vanilla	pointer	-3.333×10^{-5}	$[-1.71 \times 10^{-4}, 1.13 \times 10^{-4}]$	0.9659	-0.021	0.8812
vanilla	seq2seq	-0.002877	[-0.00425, -0.0015]	0.0009	-0.467	0.0006

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.005767	[-0.00952, 0.0211]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.007433	[-0.0173, 0.0322]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.005467	[-0.0173, 0.0282]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.005867	[-0.00954, 0.0213]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.0049	[-0.0172, 0.027]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.003	[-0.0156, 0.0216]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0039	[-0.0118, 0.0196]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.005367	[-0.00944, 0.0202]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.02533	[0.00884, 0.0418]	0.1690
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0275	[0.0129, 0.0421]	0.1005
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.02963	[0.012, 0.0473]	0.1348
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.03253	[0.0153, 0.0497]	0.1005
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.0167	[-0.00318, 0.0366]	0.9605
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.005933	[-0.0284, 0.0403]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.003933	[-0.0161, 0.0239]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.003833	[-0.0162, 0.0239]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.007667	[-0.00733, 0.0227]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0048	[-0.0121, 0.0217]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.004067	[-0.0115, 0.0196]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0045	[-0.0105, 0.0195]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.004333	[-0.00992, 0.0186]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0047	[-0.0113, 0.0207]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.004833	[-0.0106, 0.0202]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0065	[-0.00839, 0.0214]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01167	$[1.51 \times 10^{-5}, 0.0233]$	0.7472
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.01187	[0.00154, 0.0222]	0.5317
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.01027	[-0.00417, 0.0247]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.008167	[-0.00244, 0.0188]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01277	[-0.00966, 0.0352]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0016	[-0.0217, 0.0185]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.001033	[-0.0191, 0.017]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.001133	[-0.0193, 0.017]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.007433	[-0.00142, 0.0163]	0.8805
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0079	[-0.00324, 0.019]	0.9521
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0071	[-0.00237, 0.0166]	0.9521
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01183	$[7.26 \times 10^{-4}, 0.0229]$	0.5493

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.005033	[-0.00916, 0.0192]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.0219	[0.00175, 0.042]	0.5493
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.03633	[0.0196, 0.0531]	0.0793
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.03767	[0.0212, 0.0541]	0.0722
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.008133	[-0.0322, 0.0159]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0051	[-0.0274, 0.0172]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01087	[-0.023, 0.00128]	0.8152
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01117	[-0.0305, 0.0082]	0.9521
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.001667	[-0.0222, 0.0255]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01267	[-0.00607, 0.0314]	0.9521
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.03043	[0.00656, 0.0543]	0.4085
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.03157	[0.0119, 0.0512]	0.2127
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01143	[-0.00223, 0.0251]	0.8805
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01497	[0.00235, 0.0276]	0.4519
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0133	[0.00264, 0.024]	0.4115
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.0155	[0.00498, 0.026]	0.2693
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01347	[-0.00462, 0.0316]	0.9521
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.0156	[0.00843, 0.0228]	0.0793
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.03047	[0.0171, 0.0438]	0.0722
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.03003	[0.0182, 0.0419]	0.0521

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.279	[0.26, 0.298]	0.0253	5	0
adaptive	pointer	0.01	0.2589	[0.23, 0.279]	0.032	5	0
adaptive	pointer	0.1	0.2734	[0.262, 0.284]	0.0143	5	0
adaptive	pointer	1	0.2704	[0.238, 0.29]	0.0362	5	0
adaptive	pointer	10	0.2713	[0.259, 0.282]	0.0153	5	0
adaptive	pointer	100	0.2647	[0.232, 0.283]	0.0367	5	0
adaptive	pointer	1000	0.2708	[0.236, 0.292]	0.0384	5	0
adaptive	pointer	1e+04	0.2672	[0.228, 0.291]	0.0437	5	0

Table 9: ood_val_f1 per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.09216	[0.0596, 0.125]	0.0435	5	0
adaptive	seq2seq	0.01	0.1144	[0.0655, 0.156]	0.0594	5	0
adaptive	seq2seq	0.1	0.1114	[0.0528, 0.156]	0.066	5	0
adaptive	seq2seq	1	0.09909	[0.0764, 0.124]	0.0323	5	0
adaptive	seq2seq	10	0.07335	[0.0251, 0.123]	0.0656	5	0
adaptive	seq2seq	100	0.06491	[0.0179, 0.112]	0.059	5	0
adaptive	seq2seq	1000	0	[0, 0]	0	5	0
adaptive	seq2seq	1e+04	0	[0, 0]	0	5	0
tropical	pointer	0	0.1807	[0.107, 0.248]	0.0907	5	0
tropical	pointer	0.01	0.3131	[0.214, 0.418]	0.137	5	0
tropical	pointer	0.1	0.2183	[0.14, 0.306]	0.107	5	0
tropical	pointer	1	0.1911	[0.101, 0.305]	0.129	5	0
tropical	pointer	10	0.2338	[0.166, 0.309]	0.0915	5	0
tropical	pointer	100	0.2181	[0.122, 0.314]	0.126	5	0
tropical	pointer	1000	0.2155	[0.161, 0.271]	0.0712	5	0
tropical	pointer	1e+04	0.235	[0.142, 0.326]	0.12	5	0
tropical	seq2seq	0	0.04794	$[1.13 \times 10^{-5}, 0.13]$	0.0896	5	0
tropical	seq2seq	0.01	0.1057	[0.0389, 0.173]	0.0903	5	0
tropical	seq2seq	0.1	0.08684	[0.00696, 0.169]	0.104	5	0
tropical	seq2seq	1	0.05932	[0.005, 0.129]	0.0792	5	0
tropical	seq2seq	10	0.127	[0.043, 0.211]	0.106	5	0
tropical	seq2seq	100	0.007478	[0, 0.0223]	0.0166	5	0
tropical	seq2seq	1000	0.01068	[0, 0.032]	0.0239	5	0
tropical	seq2seq	1e+04	0.01068	[0, 0.032]	0.0239	5	0
vanilla	pointer	0	0.2401	[0.146, 0.294]	0.105	5	0
vanilla	pointer	0.01	0.2443	[0.183, 0.303]	0.0791	5	0
vanilla	pointer	0.1	0.2461	[0.183, 0.307]	0.0815	5	0
vanilla	pointer	1	0.2376	[0.17, 0.305]	0.0856	5	0
vanilla	pointer	10	0.256	[0.203, 0.302]	0.0647	5	0
vanilla	pointer	100	0.2524	[0.2, 0.302]	0.0681	5	0
vanilla	pointer	1000	0.2559	[0.204, 0.308]	0.0685	5	0
vanilla	pointer	1e+04	0.2461	[0.176, 0.313]	0.089	5	0
vanilla	seq2seq	0	0.03673	[0.00836, 0.0637]	0.0356	5	0
vanilla	seq2seq	0.01	0.02722	[0.00763, 0.0542]	0.0315	5	0
vanilla	seq2seq	0.1	0.02768	[0.00215, 0.0532]	0.0345	5	0
vanilla	seq2seq	1	0.06688	[0.00514, 0.13]	0.0832	5	0
vanilla	seq2seq	10	0.04036	[0.0163, 0.0643]	0.0314	5	0
vanilla	seq2seq	100	0.1109	[0.0759, 0.143]	0.0425	5	0
vanilla	seq2seq	1000	0	[0, 0]	0	5	0
vanilla	seq2seq	1e+04	0	[0, 0]	0	5	0

Table 10: ood_val_f1: each level of fishr_lambda against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.007539	[-0.065, 0.0106]	1.0000
adaptive	pointer	0.1	0.005768	[-0.0316, 0.0171]	1.0000
adaptive	pointer	1	0.009014	[-0.0575, 0.0253]	1.0000
adaptive	pointer	10	0.001851	[-0.0365, 0.0158]	1.0000
adaptive	pointer	100	-0.00694	[-0.0635, 0.0209]	1.0000
adaptive	pointer	1000	0.008859	[-0.0586, 0.0254]	1.0000
adaptive	pointer	1e+04	-1.281×10^{-4}	[-0.0687, 0.0256]	1.0000
adaptive	seq2seq	0.01	-0.0015	[-0.0152, 0.0596]	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	0.1	0.02819	[-0.0131, 0.0458]	1.0000
adaptive	seq2seq	1	-0.01279	[-0.0466, 0.0605]	1.0000
adaptive	seq2seq	10	-0.009937	[-0.0559, 0.0182]	1.0000
adaptive	seq2seq	100	-0.03244	[-0.0823, 0.0273]	1.0000
adaptive	seq2seq	1000	-0.09957	[-0.125, -0.0603]	0.4375
adaptive	seq2seq	1e+04	-0.09957	[-0.125, -0.0603]	0.4375
tropical	pointer	0.01	0.129	[-0.00704, 0.272]	1.0000
tropical	pointer	0.1	0.0294	[-0.0982, 0.188]	1.0000
tropical	pointer	1	0.007073	[-0.0792, 0.0923]	1.0000
tropical	pointer	10	0.06995	[-0.00324, 0.109]	1.0000
tropical	pointer	100	0.07303	[-0.0522, 0.11]	1.0000
tropical	pointer	1000	0.03297	[-0.0689, 0.148]	1.0000
tropical	pointer	1e+04	0.01755	[-0.0449, 0.166]	1.0000
tropical	seq2seq	0.01	0.02838	$[7.43 \times 10^{-4}, 0.118]$	1.0000
tropical	seq2seq	0.1	0	[-0.0141, 0.117]	1.0000
tropical	seq2seq	1	3.262×10^{-4}	[-0.067, 0.105]	1.0000
tropical	seq2seq	10	0.03551	[0.0052, 0.158]	1.0000
tropical	seq2seq	100	-5.632×10^{-5}	[-0.124, 0.0149]	1.0000
tropical	seq2seq	1000	-5.632×10^{-5}	[-0.124, 0.0215]	1.0000
tropical	seq2seq	1e+04	-5.632×10^{-5}	[-0.124, 0.0215]	1.0000
vanilla	pointer	0.01	0.007002	[-0.0645, 0.0629]	1.0000
vanilla	pointer	0.1	0.008204	[-0.0582, 0.0587]	1.0000
vanilla	pointer	1	0.01309	[-0.0702, 0.0513]	1.0000
vanilla	pointer	10	0.0051	[-0.0657, 0.1]	1.0000
vanilla	pointer	100	0.006125	[-0.067, 0.091]	1.0000
vanilla	pointer	1000	0.008169	[-0.0666, 0.0977]	1.0000
vanilla	pointer	1e+04	0.02306	[-0.0603, 0.0566]	1.0000
vanilla	seq2seq	0.01	3.889×10^{-4}	[-0.0339, 0.0149]	1.0000
vanilla	seq2seq	0.1	-0.003456	$[-0.0233, 4.46 \times 10^{-4}]$	1.0000
vanilla	seq2seq	1	0.01133	[-0.0114, 0.0717]	1.0000
vanilla	seq2seq	10	0.001111	[-0.00664, 0.0147]	1.0000
vanilla	seq2seq	100	0.08463	[0.0408, 0.107]	0.4375
vanilla	seq2seq	1000	-0.04122	[-0.0637, -0.00836]	0.7500
vanilla	seq2seq	1e+04	-0.04122	[-0.0637, -0.00836]	0.7500

Table 11: `ood_val_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.2037	[-0.194, 0.519]	0.2405	0.158	0.2370
adaptive	seq2seq	-0.7392	[-0.862, -0.636]	$< 10^{-4}$	-0.589	0.0002
tropical	pointer	-0.1061	[-0.478, 0.229]	0.5442	-0.0781	0.5591
tropical	seq2seq	-0.4387	[-0.707, -0.165]	0.0084	-0.32	0.0096
vanilla	pointer	0.03961	[-0.156, 0.137]	0.8213	0.0324	0.8196
vanilla	seq2seq	-0.4716	[-0.542, -0.405]	0.0042	-0.312	0.0150

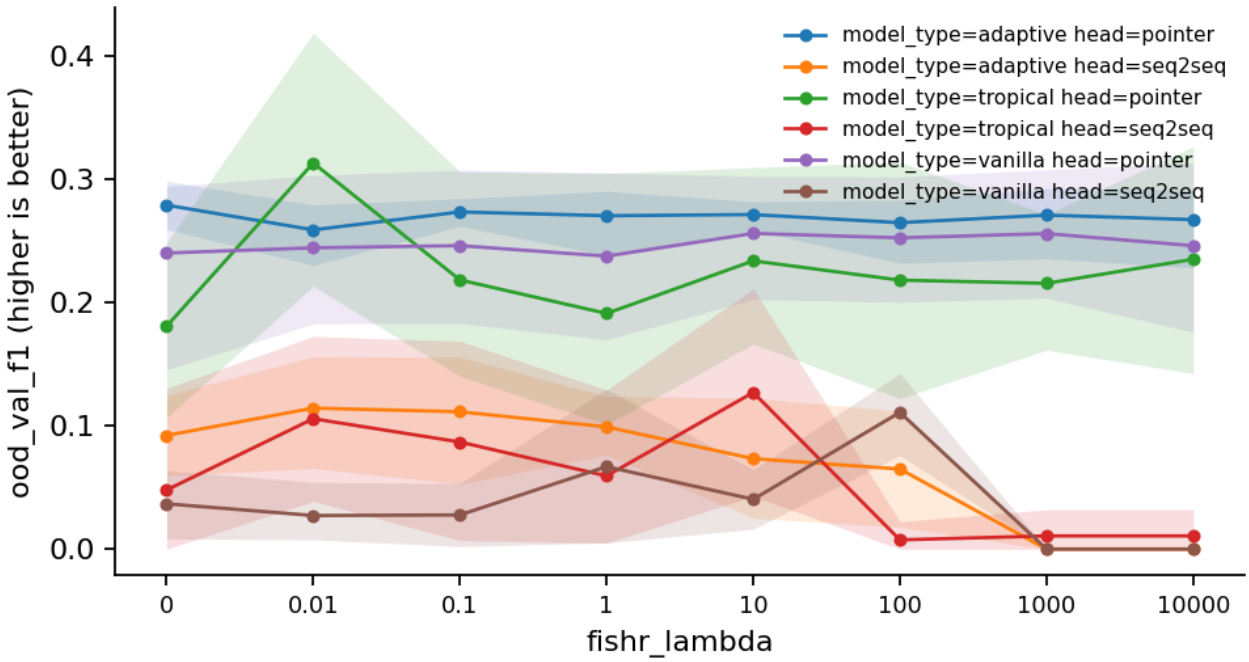


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.05936	[-0.271, 0.152]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06882	[-0.156, 0.294]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.02781	[-0.172, 0.116]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.04649	[-0.304, 0.211]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.02226	[-0.21, 0.165]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.03435	[-0.259, 0.19]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.04033	[-0.152, 0.0709]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.01102	[-0.217, 0.195]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.01121	[-0.0907, 0.113]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.07849	[-0.0621, 0.219]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.05916	[-0.0643, 0.183]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.007553	[-0.163, 0.148]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.08665	[-0.0368, 0.21]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.1034	[-0.154, -0.0524]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.01068	[-0.019, 0.0403]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.01068	[-0.019, 0.0403]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.03887	[-0.0763, 0.154]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01456	[-0.0709, 0.1]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02733	[-0.0707, 0.125]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0328	[-0.0536, 0.119]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01529	[-0.0523, 0.0828]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.01227	[-0.0485, 0.073]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.015	[-0.0416, 0.0716]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02114	[-0.0676, 0.11]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.05544	[-0.0133, 0.124]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.08714	[0.0111, 0.163]	0.8750
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.08367	[0.0132, 0.154]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.03222	[-0.0757, 0.14]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.033	[-0.044, 0.11]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.04595	[-0.164, 0.0724]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0	[–, –]	–
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0	[–, –]	–
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1868	[0.133, 0.241]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1445	[0.0564, 0.233]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1621	[0.0762, 0.248]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1713	[0.114, 0.229]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1979	[0.118, 0.278]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1998	[0.0945, 0.305]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2708	[0.223, 0.319]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.2672	[0.213, 0.321]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1328	[-0.055, 0.321]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.2074	[-0.0656, 0.48]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1314	[-0.074, 0.337]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1318	[-0.0542, 0.318]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1068	[-0.078, 0.292]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.2106	[0.0674, 0.354]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2049	[0.128, 0.282]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.2244	[0.0721, 0.377]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.2034	[0.0665, 0.34]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.2171	[0.123, 0.311]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.2184	[0.145, 0.291]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1707	[0.117, 0.225]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.2157	[0.142, 0.29]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1416	[0.0607, 0.223]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.2559	[0.171, 0.341]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2461	[0.136, 0.357]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.1626	[0.109, 0.254]	0.102	5	0
adaptive	pointer	0.01	0.1218	[0.109, 0.136]	0.0172	5	0
adaptive	pointer	0.1	0.1217	[0.113, 0.133]	0.0134	5	0
adaptive	pointer	1	0.1652	[0.12, 0.245]	0.0878	5	0
adaptive	pointer	10	0.1578	[0.116, 0.238]	0.0895	5	0
adaptive	pointer	100	0.1209	[0.118, 0.124]	0.00429	5	0
adaptive	pointer	1000	0.1235	[0.113, 0.137]	0.0151	5	0
adaptive	pointer	1e+04	0.138	[0.126, 0.15]	0.0159	5	0
adaptive	seq2seq	0	0.07734	[0.0528, 0.11]	0.0369	5	0
adaptive	seq2seq	0.01	0.08818	[0.0689, 0.108]	0.0242	5	0
adaptive	seq2seq	0.1	0.08374	[0.0611, 0.105]	0.0291	5	0
adaptive	seq2seq	1	0.0889	[0.0657, 0.117]	0.0341	5	0
adaptive	seq2seq	10	0.1484	[0.064, 0.299]	0.167	5	0
adaptive	seq2seq	100	0.1448	[0.0543, 0.312]	0.183	5	0
adaptive	seq2seq	1000	0.4928	[0.472, 0.506]	0.0233	5	0
adaptive	seq2seq	1e+04	0.4928	[0.472, 0.506]	0.0233	5	0
tropical	pointer	0	0.3119	[0.275, 0.343]	0.0442	5	0
tropical	pointer	0.01	0.3735	[0.325, 0.445]	0.0809	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.3447	[0.294, 0.392]	0.0659	5	0
tropical	pointer	1	0.3023	[0.269, 0.348]	0.0533	5	0
tropical	pointer	10	0.3392	[0.281, 0.401]	0.0783	5	0
tropical	pointer	100	0.3286	[0.29, 0.367]	0.0503	5	0
tropical	pointer	1000	0.3111	[0.286, 0.335]	0.0324	5	0
tropical	pointer	1e+04	0.3498	[0.298, 0.397]	0.0626	5	0
tropical	seq2seq	0	0.05946	[0.0448, 0.0742]	0.0199	5	0
tropical	seq2seq	0.01	0.1114	[0.0897, 0.151]	0.0441	5	0
tropical	seq2seq	0.1	0.06698	[0.051, 0.0835]	0.0217	5	0
tropical	seq2seq	1	0.07906	[0.0593, 0.102]	0.0288	5	0
tropical	seq2seq	10	0.0856	[0.0674, 0.0983]	0.02	5	0
tropical	seq2seq	100	0.1428	[0.0524, 0.274]	0.146	5	0
tropical	seq2seq	1000	0.353	[0.207, 0.475]	0.177	5	0
tropical	seq2seq	1e+04	0.353	[0.204, 0.479]	0.177	5	0
vanilla	pointer	0	0.1449	[0.116, 0.174]	0.0375	5	0
vanilla	pointer	0.01	0.1898	[0.145, 0.235]	0.0615	5	0
vanilla	pointer	0.1	0.194	[0.146, 0.247]	0.0659	5	0
vanilla	pointer	1	0.1867	[0.138, 0.238]	0.0667	5	0
vanilla	pointer	10	0.2164	[0.156, 0.278]	0.0781	5	0
vanilla	pointer	100	0.2057	[0.151, 0.26]	0.07	5	0
vanilla	pointer	1000	0.215	[0.149, 0.277]	0.0787	5	0
vanilla	pointer	1e+04	0.1857	[0.145, 0.239]	0.0602	5	0
vanilla	seq2seq	0	0.05556	[0.0445, 0.0694]	0.0161	5	0
vanilla	seq2seq	0.01	0.05326	[0.0447, 0.0668]	0.0145	5	0
vanilla	seq2seq	0.1	0.05344	[0.0464, 0.0633]	0.0113	5	0
vanilla	seq2seq	1	0.06728	[0.0516, 0.084]	0.0207	5	0
vanilla	seq2seq	10	0.06936	[0.0519, 0.0888]	0.0234	5	0
vanilla	seq2seq	100	0.07498	[0.0599, 0.0901]	0.0193	5	0
vanilla	seq2seq	1000	0.5012	[0.496, 0.506]	0.00623	5	0
vanilla	seq2seq	1e+04	0.5012	[0.496, 0.506]	0.00623	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.04086	[-0.142, 0.0607]	0.7875	
adaptive	pointer	0.1	-0.04094	[-0.142, 0.0606]	0.7863	
adaptive	pointer	1	0.00254	[-0.099, 0.104]	1.0000	
adaptive	pointer	10	-0.00488	[-0.106, 0.0966]	1.0000	
adaptive	pointer	100	-0.04174	[-0.143, 0.0598]	0.7720	
adaptive	pointer	1000	-0.03912	[-0.141, 0.0624]	0.8171	
adaptive	pointer	1e+04	-0.02466	[-0.126, 0.0769]	0.9761	
adaptive	seq2seq	0.01	0.01084	[-0.148, 0.17]	1.0000	
adaptive	seq2seq	0.1	0.0064	[-0.153, 0.166]	1.0000	
adaptive	seq2seq	1	0.01156	[-0.148, 0.171]	1.0000	
adaptive	seq2seq	10	0.07102	[-0.0881, 0.23]	0.7061	
adaptive	seq2seq	100	0.06748	[-0.0917, 0.227]	0.7483	
adaptive	seq2seq	1000	0.4154	[0.256, 0.575]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	0.4154	[0.256, 0.575]	$< 10^{-4}$	*
tropical	pointer	0.01	0.0616	[-0.0441, 0.167]	0.4461	
tropical	pointer	0.1	0.03284	[-0.0729, 0.139]	0.9223	
tropical	pointer	1	-0.00952	[-0.115, 0.0962]	0.9999	
tropical	pointer	10	0.02734	[-0.0784, 0.133]	0.9672	
tropical	pointer	100	0.01676	[-0.089, 0.123]	0.9979	

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	pointer	1000	-7.800×10^{-4}	[-0.107, 0.105]	1.0000	
tropical	pointer	1e+04	0.03796	[-0.0678, 0.144]	0.8592	
tropical	seq2seq	0.01	0.05196	[-0.131, 0.235]	0.9481	
tropical	seq2seq	0.1	0.00752	[-0.175, 0.19]	1.0000	
tropical	seq2seq	1	0.0196	[-0.163, 0.203]	0.9998	
tropical	seq2seq	10	0.02614	[-0.157, 0.209]	0.9989	
tropical	seq2seq	100	0.08336	[-0.0995, 0.266]	0.6878	
tropical	seq2seq	1000	0.2935	[0.111, 0.476]	0.0006	*
tropical	seq2seq	1e+04	0.2935	[0.111, 0.476]	0.0007	*
vanilla	pointer	0.01	0.04494	[-0.0703, 0.16]	0.8090	
vanilla	pointer	0.1	0.04906	[-0.0661, 0.164]	0.7451	
vanilla	pointer	1	0.0418	[-0.0734, 0.157]	0.8533	
vanilla	pointer	10	0.07152	[-0.0437, 0.187]	0.3813	
vanilla	pointer	100	0.06076	[-0.0544, 0.176]	0.5479	
vanilla	pointer	1000	0.07014	[-0.0451, 0.185]	0.4009	
vanilla	pointer	1e+04	0.04084	[-0.0744, 0.156]	0.8658	
vanilla	seq2seq	0.01	-0.0023	[-0.0301, 0.0255]	1.0000	
vanilla	seq2seq	0.1	-0.00212	[-0.0299, 0.0257]	1.0000	
vanilla	seq2seq	1	0.01172	[-0.0161, 0.0395]	0.7523	
vanilla	seq2seq	10	0.0138	[-0.014, 0.0416]	0.6074	
vanilla	seq2seq	100	0.01942	[-0.00836, 0.0472]	0.2680	
vanilla	seq2seq	1000	0.4457	[0.418, 0.473]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	0.4457	[0.418, 0.473]	$< 10^{-4}$	*

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.843×10^{-4}	[-0.0031, 0.00386]	0.9451	0.158	0.2388
adaptive	seq2seq	0.07456	[0.07, 0.079]	$< 10^{-4}$	0.419	0.0012
tropical	pointer	-0.003996	[-0.0132, 0.00524]	0.4472	-0.0705	0.5921
tropical	seq2seq	0.04859	[0.0221, 0.069]	$< 10^{-4}$	0.322	0.0162
vanilla	pointer	0.001744	$[-9.83 \times 10^{-4}$, 0.00543]	0.7520	-0.00952	0.9536
vanilla	seq2seq	0.08025	[0.0779, 0.0819]	$< 10^{-4}$	0.68	0.0002

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.167	[0.0867, 0.247]	0.0581	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.1836	[0.115, 0.252]	0.0284	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1507	[0.0427, 0.259]	0.1973	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1156	[0.0693, 0.162]	0.0341	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1228	[0.0433, 0.202]	0.1532	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.123	[0.0303, 0.216]	0.2116	

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.09604	[-0.0254, 0.217]	0.7448
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1641	[0.0871, 0.241]	0.0572
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.0039	[-0.0192, 0.027]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.05816	[-0.00486, 0.121]	0.5624
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01354	[-0.0198, 0.0468]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01178	[-0.0329, 0.0565]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01624	[-0.0126, 0.0451]	0.9707
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.06784	[-0.12, 0.256]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.1482	[-0.371, 0.0747]	0.9707
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.1482	[-0.371, 0.0747]	0.9707
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01774	[-0.0775, 0.113]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.06806	[-0.125, -0.0115]	0.4318
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.07226	[-0.139, -0.00576]	0.5443
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.02152	[-0.0863, 0.0433]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.05866	[-0.172, 0.0546]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.08476	[-0.172, 0.00236]	0.5942
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.09152	[-0.176, -0.00663]	0.5443
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.04776	[-0.116, 0.0208]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.02178	[-0.0122, 0.0558]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.03492	[0.0104, 0.0595]	0.2686
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.0303	[0.00251, 0.0581]	0.5443
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.02162	[-0.00791, 0.0511]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.079	[-0.116, 0.274]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.06984	[-0.146, 0.286]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00844	[-0.0319, 0.015]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00844	[-0.0319, 0.015]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.0853	[-0.0397, 0.21]	0.7860
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0336	[-0.00218, 0.0694]	0.4173

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03796	[6.05×10^{-4}, 0.0753]	0.3821	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.07628	[-0.0438, 0.196]	0.7860	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.0094	[-0.247, 0.266]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02392	[-0.254, 0.206]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.3693	[-0.402, -0.336]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.3548	[-0.399, -0.311]	0.0005	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.2524	[0.174, 0.331]	0.0177	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.262	[0.211, 0.313]	0.0030	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.2777	[0.18, 0.376]	0.0240	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.2233	[0.151, 0.295]	0.0192	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2536	[0.164, 0.344]	0.0240	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.1858	[0.0316, 0.34]	0.2581	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.0419	[-0.288, 0.204]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.00316	[-0.244, 0.238]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.08934	[0.0418, 0.137]	0.0969	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1366	[0.0537, 0.22]	0.1331	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1405	[0.054, 0.227]	0.1331	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1194	[0.0258, 0.213]	0.2396	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1471	[0.0597, 0.234]	0.1326	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1307	[0.0484, 0.213]	0.1331	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.2862	[-0.38, -0.192]	0.0195	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.3155	[-0.392, -0.239]	0.0071	*

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.3018	[0.296, 0.309]	0.00889	5	0
adaptive	pointer	0.01	0.2993	[0.295, 0.303]	0.00521	5	0
adaptive	pointer	0.1	0.3018	[0.294, 0.308]	0.00852	5	0
adaptive	pointer	1	0.3029	[0.294, 0.308]	0.00935	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.3001	[0.292, 0.307]	0.00995	5	0
adaptive	pointer	100	0.3024	[0.298, 0.307]	0.00642	5	0
adaptive	pointer	1000	0.2999	[0.293, 0.306]	0.00854	5	0
adaptive	pointer	1e+04	0.3019	[0.296, 0.308]	0.00756	5	0
adaptive	seq2seq	0	0.2338	[0.216, 0.259]	0.0289	5	0
adaptive	seq2seq	0.01	0.2273	[0.203, 0.252]	0.0317	5	0
adaptive	seq2seq	0.1	0.2361	[0.207, 0.267]	0.0399	5	0
adaptive	seq2seq	1	0.2387	[0.217, 0.26]	0.028	5	0
adaptive	seq2seq	10	0.1972	[0.181, 0.214]	0.0194	5	0
adaptive	seq2seq	100	0.1697	[0.15, 0.189]	0.0246	5	0
adaptive	seq2seq	1000	0.09155	[0.0449, 0.136]	0.0613	5	0
adaptive	seq2seq	1e+04	0.09155	[0.0436, 0.136]	0.0613	5	0
tropical	pointer	0	0.3206	[0.311, 0.333]	0.0136	5	0
tropical	pointer	0.01	0.3256	[0.312, 0.338]	0.0157	5	0
tropical	pointer	0.1	0.3262	[0.317, 0.334]	0.0114	5	0
tropical	pointer	1	0.3251	[0.313, 0.333]	0.0139	5	0
tropical	pointer	10	0.3216	[0.308, 0.333]	0.0158	5	0
tropical	pointer	100	0.3235	[0.315, 0.331]	0.0101	5	0
tropical	pointer	1000	0.3214	[0.313, 0.329]	0.0106	5	0
tropical	pointer	1e+04	0.3197	[0.309, 0.328]	0.0122	5	0
tropical	seq2seq	0	0.2669	[0.227, 0.307]	0.0516	5	0
tropical	seq2seq	0.01	0.2995	[0.251, 0.334]	0.0562	5	0
tropical	seq2seq	0.1	0.3054	[0.268, 0.343]	0.0482	5	0
tropical	seq2seq	1	0.2762	[0.222, 0.33]	0.07	5	0
tropical	seq2seq	10	0.2762	[0.239, 0.312]	0.0482	5	0
tropical	seq2seq	100	0.2083	[0.136, 0.289]	0.0952	5	0
tropical	seq2seq	1000	0.1331	[0.0956, 0.185]	0.0598	5	0
tropical	seq2seq	1e+04	0.1331	[0.0957, 0.185]	0.0598	5	0
vanilla	pointer	0	0.2839	[0.265, 0.304]	0.0248	5	0
vanilla	pointer	0.01	0.275	[0.257, 0.293]	0.0231	5	0
vanilla	pointer	0.1	0.2773	[0.26, 0.294]	0.0231	5	0
vanilla	pointer	1	0.2763	[0.26, 0.292]	0.0218	5	0
vanilla	pointer	10	0.2768	[0.26, 0.293]	0.0217	5	0
vanilla	pointer	100	0.2758	[0.259, 0.293]	0.0235	5	0
vanilla	pointer	1000	0.2771	[0.259, 0.296]	0.0241	5	0
vanilla	pointer	1e+04	0.2776	[0.259, 0.297]	0.0248	5	0
vanilla	seq2seq	0	0.2191	[0.205, 0.236]	0.0214	5	0
vanilla	seq2seq	0.01	0.2059	[0.197, 0.215]	0.0118	5	0
vanilla	seq2seq	0.1	0.2022	[0.197, 0.208]	0.00666	5	0
vanilla	seq2seq	1	0.2078	[0.195, 0.228]	0.0218	5	0
vanilla	seq2seq	10	0.2104	[0.2, 0.223]	0.0149	5	0
vanilla	seq2seq	100	0.172	[0.157, 0.186]	0.0185	5	0
vanilla	seq2seq	1000	0.1103	[0.0808, 0.149]	0.044	5	0
vanilla	seq2seq	1e+04	0.1106	[0.0809, 0.148]	0.0437	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.002487	[-0.0168, 0.0118]	0.9964
adaptive	pointer	0.1	2.962×10^{-5}	[-0.0143, 0.0143]	1.0000
adaptive	pointer	1	0.001109	[-0.0132, 0.0154]	1.0000
adaptive	pointer	10	-0.001728	[-0.016, 0.0126]	0.9996
adaptive	pointer	100	6.115×10^{-4}	[-0.0137, 0.0149]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	1000	-0.001907	[-0.0162, 0.0124]	0.9993	
adaptive	pointer	1e+04	1.639×10^{-4}	[-0.0141, 0.0145]	1.0000	
adaptive	seq2seq	0.01	-0.006479	[-0.0761, 0.0631]	0.9999	
adaptive	seq2seq	0.1	0.002361	[-0.0673, 0.072]	1.0000	
adaptive	seq2seq	1	0.004944	[-0.0647, 0.0746]	1.0000	
adaptive	seq2seq	10	-0.03655	[-0.106, 0.0331]	0.5526	
adaptive	seq2seq	100	-0.06409	[-0.134, 0.00553]	0.0813	
adaptive	seq2seq	1000	-0.1422	[-0.212, -0.0726]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	-0.1422	[-0.212, -0.0726]	$< 10^{-4}$	*
tropical	pointer	0.01	0.005024	[-0.0178, 0.0279]	0.9860	
tropical	pointer	0.1	0.005645	[-0.0172, 0.0285]	0.9739	
tropical	pointer	1	0.004544	[-0.0183, 0.0274]	0.9920	
tropical	pointer	10	0.001028	[-0.0218, 0.0239]	1.0000	
tropical	pointer	100	0.002901	[-0.0199, 0.0257]	0.9995	
tropical	pointer	1000	8.828×10^{-4}	[-0.022, 0.0237]	1.0000	
tropical	pointer	1e+04	-8.864×10^{-4}	[-0.0237, 0.022]	1.0000	
tropical	seq2seq	0.01	0.03255	[-0.0772, 0.142]	0.9367	
tropical	seq2seq	0.1	0.03843	[-0.0713, 0.148]	0.8722	
tropical	seq2seq	1	0.009271	[-0.1, 0.119]	1.0000	
tropical	seq2seq	10	0.009266	[-0.1, 0.119]	1.0000	
tropical	seq2seq	100	-0.05862	[-0.168, 0.0511]	0.5350	
tropical	seq2seq	1000	-0.1339	[-0.244, -0.0241]	0.0115	*
tropical	seq2seq	1e+04	-0.1339	[-0.244, -0.0241]	0.0116	*
vanilla	pointer	0.01	-0.008953	[-0.0498, 0.0319]	0.9863	
vanilla	pointer	0.1	-0.006635	[-0.0475, 0.0343]	0.9976	
vanilla	pointer	1	-0.007635	[-0.0485, 0.0333]	0.9944	
vanilla	pointer	10	-0.007142	[-0.048, 0.0338]	0.9963	
vanilla	pointer	100	-0.008125	[-0.049, 0.0328]	0.9920	
vanilla	pointer	1000	-0.006896	[-0.0478, 0.034]	0.9970	
vanilla	pointer	1e+04	-0.006344	[-0.0472, 0.0346]	0.9982	
vanilla	seq2seq	0.01	-0.01322	[-0.0592, 0.0327]	0.9449	
vanilla	seq2seq	0.1	-0.01695	[-0.0629, 0.029]	0.8437	
vanilla	seq2seq	1	-0.01137	[-0.0573, 0.0346]	0.9737	
vanilla	seq2seq	10	-0.008713	[-0.0547, 0.0372]	0.9939	
vanilla	seq2seq	100	-0.04716	[-0.0931, -0.00122]	0.0422	*
vanilla	seq2seq	1000	-0.1088	[-0.155, -0.0629]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	-0.1086	[-0.155, -0.0626]	$< 10^{-4}$	*

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.279×10^{-4}	$[-2.69 \times 10^{-4}, 5.79 \times 10^{-4}]$	0.8426	0.0552	0.6965
adaptive	seq2seq	-0.02733	[-0.0397, -0.015]	$< 10^{-4}$	-0.716	0.0002
tropical	pointer	-0.001032	$[-0.00177, -2.35 \times 10^{-4}]$	0.3169	-0.135	0.3177
tropical	seq2seq	-0.03256	[-0.0405, -0.0245]	$< 10^{-4}$	-0.547	0.0002
vanilla	pointer	2.433×10^{-4}	$[-1.14 \times 10^{-4}, 6.83 \times 10^{-4}]$	0.8937	-0.0019	1.0000
vanilla	seq2seq	-0.01805	[-0.0244, -0.0105]	$< 10^{-4}$	-0.577	0.0002

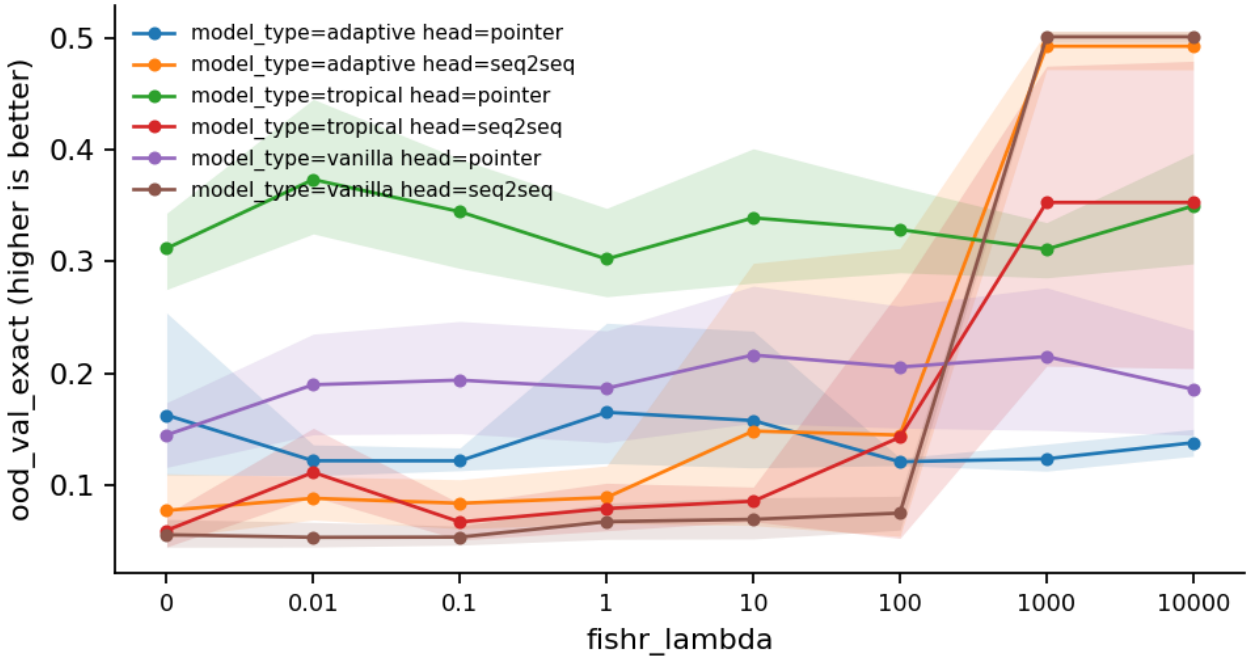


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

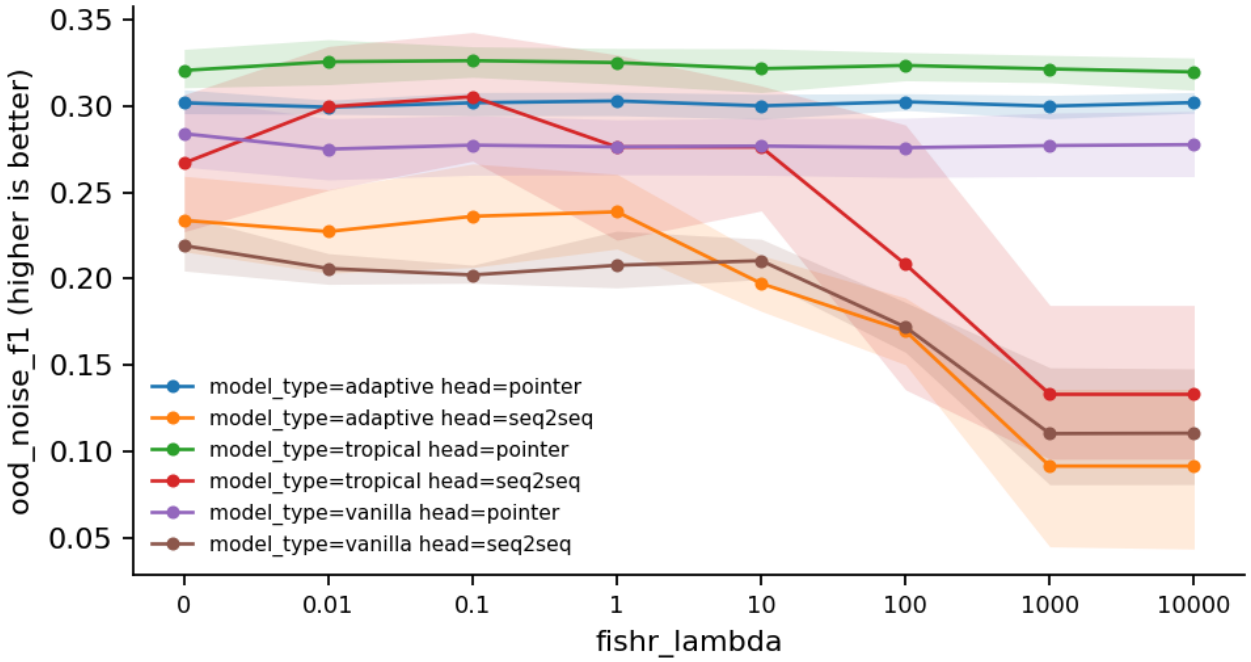


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0366	[-0.00401, 0.0772]	0.3995
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.05058	[0.00765, 0.0935]	0.3987
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.04888	[0.00841, 0.0894]	0.3987
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.04878	[0.00794, 0.0896]	0.3987
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.04477	[0.00523, 0.0843]	0.3987
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04763	[0.0095, 0.0858]	0.3843
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.04438	[0.00605, 0.0827]	0.3987
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.04206	[0.00156, 0.0826]	0.3987
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.04781	[-0.0214, 0.117]	0.5096
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.09358	[0.016, 0.171]	0.3987
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1032	[0.036, 0.17]	0.2078
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.06845	[-0.0119, 0.149]	0.3995
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.06579	[-0.00118, 0.133]	0.3987
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.03634	[-0.0817, 0.154]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.02276	[-0.0863, 0.132]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02251	[-0.0864, 0.131]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01784	[-0.0124, 0.0481]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0243	$[-1.06 \times 10^{-4}, 0.0487]$	0.5062
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0245	[0.00451, 0.0445]	0.3811
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.02658	[0.00783, 0.0453]	0.2555
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.02325	[0.00731, 0.0392]	0.2478
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02657	[0.00336, 0.0498]	0.4364
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.02282	$[9.24 \times 10^{-4}, 0.0447]$	0.4884
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02434	[0.00252, 0.0462]	0.4364
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01463	[-0.0229, 0.0522]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.02137	[-0.0082, 0.0509]	0.9223
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.03394	[-0.0141, 0.082]	0.9223
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.03094	[-0.00587, 0.0677]	0.7194

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01321	[-0.0485, 0.0221]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.002305	[-0.0473, 0.0427]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.01877	[-0.104, 0.0664]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01902	[-0.104, 0.0664]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.06803	[0.0274, 0.109]	0.1060	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.07202	[0.0321, 0.112]	0.0975	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.06569	[0.0105, 0.121]	0.2081	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.06419	[0.0215, 0.107]	0.1259	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1028	[0.0684, 0.137]	0.0266	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1327	[0.0967, 0.169]	0.0124	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2083	[0.134, 0.282]	0.0303	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.2104	[0.14, 0.281]	0.0266	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.05361	$[-9.40 \times 10^{-4}, 0.108]$	0.2825	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.02609	[-0.0401, 0.0922]	0.6700	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.02083	[-0.0345, 0.0762]	0.6700	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.04889	[-0.0326, 0.13]	0.5469	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.04537	[-0.0224, 0.113]	0.5469	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.1151	[0.0024, 0.228]	0.2825	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.1884	[0.119, 0.258]	0.0303	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1866	[0.119, 0.255]	0.0303	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.06482	[0.0148, 0.115]	0.1819	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.06909	[0.0312, 0.107]	0.0975	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.07514	[0.0482, 0.102]	0.0303	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.06855	[0.0313, 0.106]	0.0975	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.06639	[0.0262, 0.107]	0.1060	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1039	[0.0624, 0.145]	0.0382	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1667	[0.0917, 0.242]	0.0561	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.167	[0.0916, 0.243]	0.0561	

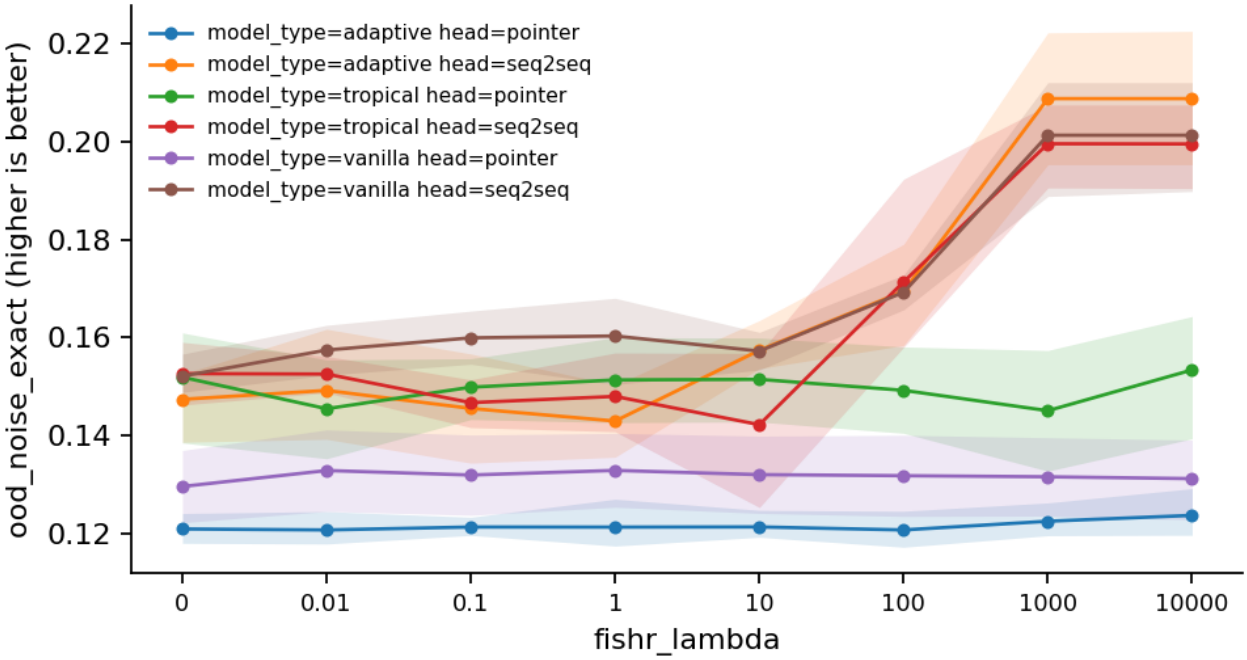


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.1209	[0.118, 0.124]	0.00404	5	0
adaptive	pointer	0.01	0.1207	[0.118, 0.124]	0.00429	5	0
adaptive	pointer	0.1	0.1213	[0.12, 0.123]	0.0025	5	0
adaptive	pointer	1	0.1213	[0.117, 0.127]	0.0062	5	0
adaptive	pointer	10	0.1213	[0.119, 0.125]	0.00376	5	0
adaptive	pointer	100	0.1207	[0.117, 0.125]	0.00491	5	0
adaptive	pointer	1000	0.1225	[0.12, 0.126]	0.00444	5	0
adaptive	pointer	1e+04	0.1237	[0.12, 0.129]	0.00636	5	0
adaptive	seq2seq	0	0.1474	[0.139, 0.153]	0.00974	5	0
adaptive	seq2seq	0.01	0.1492	[0.139, 0.162]	0.015	5	0
adaptive	seq2seq	0.1	0.1455	[0.134, 0.157]	0.0153	5	0
adaptive	seq2seq	1	0.1429	[0.136, 0.15]	0.00954	5	0
adaptive	seq2seq	10	0.1574	[0.154, 0.163]	0.00691	5	0
adaptive	seq2seq	100	0.1694	[0.158, 0.179]	0.0131	5	0
adaptive	seq2seq	1000	0.2088	[0.195, 0.222]	0.0179	5	0
adaptive	seq2seq	1e+04	0.2088	[0.195, 0.223]	0.0179	5	0
tropical	pointer	0	0.1519	[0.139, 0.161]	0.015	5	0
tropical	pointer	0.01	0.1455	[0.135, 0.155]	0.0131	5	0
tropical	pointer	0.1	0.1499	[0.143, 0.156]	0.00805	5	0
tropical	pointer	1	0.1513	[0.143, 0.16]	0.0117	5	0
tropical	pointer	10	0.1515	[0.143, 0.16]	0.0114	5	0
tropical	pointer	100	0.1492	[0.14, 0.158]	0.0119	5	0
tropical	pointer	1000	0.1451	[0.133, 0.157]	0.0157	5	0
tropical	pointer	1e+04	0.1534	[0.139, 0.164]	0.0163	5	0
tropical	seq2seq	0	0.1526	[0.146, 0.159]	0.00834	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.1525	[0.149, 0.156]	0.005	5	0
tropical	seq2seq	0.1	0.1467	[0.142, 0.151]	0.00628	5	0
tropical	seq2seq	1	0.148	[0.141, 0.157]	0.0106	5	0
tropical	seq2seq	10	0.1422	[0.125, 0.157]	0.0199	5	0
tropical	seq2seq	100	0.1714	[0.158, 0.192]	0.0232	5	0
tropical	seq2seq	1000	0.1996	[0.191, 0.208]	0.0107	5	0
tropical	seq2seq	1e+04	0.1995	[0.19, 0.208]	0.0108	5	0
vanilla	pointer	0	0.1296	[0.122, 0.137]	0.00952	5	0
vanilla	pointer	0.01	0.1328	[0.125, 0.141]	0.0107	5	0
vanilla	pointer	0.1	0.1319	[0.124, 0.14]	0.0105	5	0
vanilla	pointer	1	0.1329	[0.125, 0.14]	0.00968	5	0
vanilla	pointer	10	0.132	[0.124, 0.14]	0.0101	5	0
vanilla	pointer	100	0.1318	[0.124, 0.14]	0.0104	5	0
vanilla	pointer	1000	0.1316	[0.124, 0.14]	0.01	5	0
vanilla	pointer	1e+04	0.1312	[0.123, 0.139]	0.0108	5	0
vanilla	seq2seq	0	0.1521	[0.149, 0.157]	0.00519	5	0
vanilla	seq2seq	0.01	0.1574	[0.152, 0.163]	0.00665	5	0
vanilla	seq2seq	0.1	0.1599	[0.155, 0.165]	0.00707	5	0
vanilla	seq2seq	1	0.1603	[0.15, 0.168]	0.0118	5	0
vanilla	seq2seq	10	0.1572	[0.153, 0.161]	0.005	5	0
vanilla	seq2seq	100	0.1692	[0.166, 0.173]	0.00463	5	0
vanilla	seq2seq	1000	0.2013	[0.189, 0.212]	0.0141	5	0
vanilla	seq2seq	1e+04	0.2013	[0.19, 0.212]	0.0141	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-2.000×10^{-4}	[-0.00844, 0.00804]	1.0000	
adaptive	pointer	0.1	4.000×10^{-4}	[-0.00784, 0.00864]	1.0000	
adaptive	pointer	1	3.800×10^{-4}	[-0.00786, 0.00862]	1.0000	
adaptive	pointer	10	4.200×10^{-4}	[-0.00782, 0.00866]	1.0000	
adaptive	pointer	100	-2.000×10^{-4}	[-0.00844, 0.00804]	1.0000	
adaptive	pointer	1000	0.00158	[-0.00666, 0.00982]	0.9935	
adaptive	pointer	1e+04	0.00278	[-0.00546, 0.011]	0.8895	
adaptive	seq2seq	0.01	0.00182	[-0.0221, 0.0258]	1.0000	
adaptive	seq2seq	0.1	-0.00186	[-0.0258, 0.0221]	1.0000	
adaptive	seq2seq	1	-0.00444	[-0.0284, 0.0195]	0.9947	
adaptive	seq2seq	10	0.01002	[-0.0139, 0.034]	0.7587	
adaptive	seq2seq	100	0.02206	[-0.00189, 0.046]	0.0810	
adaptive	seq2seq	1000	0.06142	[0.0375, 0.0854]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	0.06142	[0.0375, 0.0854]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.00644	[-0.0294, 0.0165]	0.9514	
tropical	pointer	0.1	-0.00204	[-0.025, 0.0209]	1.0000	
tropical	pointer	1	-5.600×10^{-4}	[-0.0235, 0.0224]	1.0000	
tropical	pointer	10	-4.400×10^{-4}	[-0.0234, 0.0225]	1.0000	
tropical	pointer	100	-0.00268	[-0.0257, 0.0203]	0.9997	
tropical	pointer	1000	-0.00684	[-0.0298, 0.0161]	0.9358	
tropical	pointer	1e+04	0.00148	[-0.0215, 0.0245]	1.0000	
tropical	seq2seq	0.01	-1.000×10^{-4}	[-0.0233, 0.0231]	1.0000	
tropical	seq2seq	0.1	-0.00594	[-0.0292, 0.0173]	0.9690	
tropical	seq2seq	1	-0.00466	[-0.0279, 0.0186]	0.9916	
tropical	seq2seq	10	-0.01044	[-0.0337, 0.0128]	0.7004	
tropical	seq2seq	100	0.01872	[-0.00452, 0.042]	0.1562	

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
tropical	seq2seq	1000	0.04694	[0.0237, 0.0702]	$< 10^{-4}$	*
tropical	seq2seq	1e+04	0.0469	[0.0237, 0.0701]	$< 10^{-4}$	*
vanilla	pointer	0.01	0.00326	[-0.0146, 0.0211]	0.9951	
vanilla	pointer	0.1	0.00234	[-0.0155, 0.0202]	0.9994	
vanilla	pointer	1	0.00328	[-0.0146, 0.0211]	0.9949	
vanilla	pointer	10	0.00242	[-0.0154, 0.0203]	0.9992	
vanilla	pointer	100	0.0022	[-0.0156, 0.02]	0.9996	
vanilla	pointer	1000	0.00198	[-0.0159, 0.0198]	0.9998	
vanilla	pointer	1e+04	0.0016	[-0.0162, 0.0194]	0.9999	
vanilla	seq2seq	0.01	0.00532	[-0.0111, 0.0218]	0.9070	
vanilla	seq2seq	0.1	0.00782	[-0.00861, 0.0243]	0.6485	
vanilla	seq2seq	1	0.0082	[-0.00823, 0.0246]	0.6030	
vanilla	seq2seq	10	0.00512	[-0.0113, 0.0216]	0.9211	
vanilla	seq2seq	100	0.01712	$[6.90 \times 10^{-4}$, 0.0336]	0.0382	*
vanilla	seq2seq	1000	0.0492	[0.0328, 0.0656]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	0.0492	[0.0328, 0.0656]	$< 10^{-4}$	*

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	3.829×10^{-4}	$[4.36 \times 10^{-5}, 7.22 \times 10^{-4}]$	0.3187	0.149	0.2577
adaptive	seq2seq	0.01185	[0.0074, 0.0163]	$< 10^{-4}$	0.705	0.0002
tropical	pointer	4.300×10^{-4}	$[-4.55 \times 10^{-4}$, 0.00109]	0.6792	0.0724	0.6011
tropical	seq2seq	0.009648	[0.00732, 0.012]	$< 10^{-4}$	0.539	0.0004
vanilla	pointer	-2.421×10^{-4}	$[-3.68 \times 10^{-4}, -7.50 \times 10^{-5}]$	0.7650	-0.114	0.4017
vanilla	seq2seq	0.007976	[0.00568, 0.0102]	$< 10^{-4}$	0.634	0.0002

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02232	[-0.00588, 0.0505]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01262	[-0.0132, 0.0385]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01794	[-0.00301, 0.0389]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01848	[-0.00649, 0.0434]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01946	[-0.00584, 0.0448]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01744	[-0.00786, 0.0427]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0135	[-0.016, 0.043]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0222	[-0.00852, 0.0529]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	5.200×10^{-4}	[-0.0128, 0.0138]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.0049	[-0.0155, 0.00571]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.01324	[-0.0229, -0.00357]	0.3049	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.01234	[-0.0238, -8.36×10^{-4}]	0.6122	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.01504	[-0.0444, 0.0143]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.00212	[-0.0274, 0.0316]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.00174	[-0.0283, 0.0248]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.00178	[-0.0283, 0.0248]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.00868	[-0.0196, 0.00221]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.01214	[-0.0282, 0.00394]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.01062	[-0.0239, 0.00268]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.01158	[-0.0286, 0.0054]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.01068	[-0.0258, 0.00447]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.01108	[-0.0288, 0.00664]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00908	[-0.0253, 0.00714]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.0075	[-0.0262, 0.0112]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.00474	[-0.0164, 0.00695]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00824	[-0.0214, 0.00491]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01442	[-0.0392, 0.0103]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01738	[-0.0325, -0.00229]	0.5273	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	1.600×10^{-4}	[-0.0136, 0.0139]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	2.000×10^{-4}	[-0.0184, 0.0188]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00748	[-0.0229, 0.0378]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00748	[-0.0229, 0.0378]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.02648	[-0.0429, -0.0101]	0.1309	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0285	[-0.0462, -0.0108]	0.1309	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.02422	[-0.0425, -0.00592]	0.1580	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.02166	[-0.0358, -0.00752]	0.1313	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.03608	[-0.0468, -0.0254]	0.0161	*

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.04874	[-0.0636, -0.0339]	0.0161	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.08632	[-0.112, -0.0604]	0.0161	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.08512	[-0.112, -0.0587]	0.0163	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-7.400×10^{-4}	[-0.0261, 0.0246]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.00708	[-0.0197, 0.0055]	0.9663	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.00316	[-0.0107, 0.0171]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.00336	[-0.0206, 0.0274]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.00926	[-0.0259, 0.0444]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.02214	[-0.0475, 0.00327]	0.4368	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.05452	[-0.0794, -0.0296]	0.0591	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.04616	[-0.0684, -0.0239]	0.0678	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.02254	[-0.0392, -0.0059]	0.1580	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.0246	[-0.0378, -0.0114]	0.0940	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.02802	[-0.0361, -0.02]	0.0153	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.02746	[-0.0431, -0.0118]	0.1070	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.02524	[-0.0417, -0.00875]	0.1313	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.03746	[-0.0483, -0.0266]	0.0154	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.06976	[-0.0946, -0.045]	0.0261	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.07014	[-0.0972, -0.043]	0.0338	*

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4827	[0.451, 0.506]	0.0356	5	0
adaptive	pointer	0.01	0.493	[0.474, 0.507]	0.0226	5	0
adaptive	pointer	0.1	0.4935	[0.471, 0.51]	0.0242	5	0
adaptive	pointer	1	0.4959	[0.475, 0.51]	0.0235	5	0
adaptive	pointer	10	0.4927	[0.473, 0.509]	0.0226	5	0
adaptive	pointer	100	0.4929	[0.474, 0.507]	0.0223	5	0
adaptive	pointer	1000	0.4938	[0.473, 0.511]	0.024	5	0
adaptive	pointer	1e+04	0.4939	[0.472, 0.51]	0.0246	5	0
adaptive	seq2seq	0	0.3971	[0.377, 0.418]	0.0277	5	0
adaptive	seq2seq	0.01	0.3972	[0.38, 0.411]	0.0207	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.4005	[0.377, 0.418]	0.0266	5	0
adaptive	seq2seq	1	0.4087	[0.395, 0.422]	0.018	5	0
adaptive	seq2seq	10	0.3797	[0.35, 0.408]	0.038	5	0
adaptive	seq2seq	100	0.2755	[0.226, 0.326]	0.0674	5	0
adaptive	seq2seq	1000	0.194	[0.104, 0.283]	0.127	5	0
adaptive	seq2seq	1e+04	0.194	[0.0845, 0.283]	0.127	5	0
tropical	pointer	0	0.57	[0.552, 0.589]	0.0233	5	0
tropical	pointer	0.01	0.5729	[0.56, 0.59]	0.0189	5	0
tropical	pointer	0.1	0.5819	[0.572, 0.594]	0.0147	5	0
tropical	pointer	1	0.5815	[0.568, 0.595]	0.0172	5	0
tropical	pointer	10	0.5786	[0.557, 0.598]	0.0268	5	0
tropical	pointer	100	0.5804	[0.564, 0.595]	0.0201	5	0
tropical	pointer	1000	0.5678	[0.551, 0.585]	0.0216	5	0
tropical	pointer	1e+04	0.5659	[0.552, 0.581]	0.0191	5	0
tropical	seq2seq	0	0.4458	[0.377, 0.515]	0.0876	5	0
tropical	seq2seq	0.01	0.484	[0.393, 0.534]	0.101	5	0
tropical	seq2seq	0.1	0.482	[0.418, 0.541]	0.0866	5	0
tropical	seq2seq	1	0.4467	[0.369, 0.523]	0.0979	5	0
tropical	seq2seq	10	0.4399	[0.376, 0.506]	0.0858	5	0
tropical	seq2seq	100	0.3588	[0.303, 0.427]	0.0833	5	0
tropical	seq2seq	1000	0.2423	[0.214, 0.27]	0.0357	5	0
tropical	seq2seq	1e+04	0.2423	[0.214, 0.27]	0.0357	5	0
vanilla	pointer	0	0.4694	[0.443, 0.493]	0.0338	5	0
vanilla	pointer	0.01	0.4709	[0.445, 0.495]	0.0319	5	0
vanilla	pointer	0.1	0.4708	[0.443, 0.494]	0.0325	5	0
vanilla	pointer	1	0.4714	[0.445, 0.493]	0.0315	5	0
vanilla	pointer	10	0.472	[0.447, 0.494]	0.0305	5	0
vanilla	pointer	100	0.4725	[0.45, 0.495]	0.0325	5	0
vanilla	pointer	1000	0.4731	[0.447, 0.498]	0.0324	5	0
vanilla	pointer	1e+04	0.4744	[0.449, 0.495]	0.0302	5	0
vanilla	seq2seq	0	0.4207	[0.405, 0.434]	0.0185	5	0
vanilla	seq2seq	0.01	0.4207	[0.403, 0.435]	0.0205	5	0
vanilla	seq2seq	0.1	0.4196	[0.399, 0.438]	0.0249	5	0
vanilla	seq2seq	1	0.428	[0.419, 0.436]	0.0113	5	0
vanilla	seq2seq	10	0.4223	[0.405, 0.433]	0.019	5	0
vanilla	seq2seq	100	0.2907	[0.243, 0.346]	0.0654	5	0
vanilla	seq2seq	1000	0.2619	[0.22, 0.295]	0.0471	5	0
vanilla	seq2seq	1e+04	0.2621	[0.221, 0.293]	0.0467	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.01027	[-0.0339, 0.0544]	0.9809
adaptive	pointer	0.1	0.01079	[-0.0334, 0.0549]	0.9753
adaptive	pointer	1	0.01321	[-0.0309, 0.0574]	0.9341
adaptive	pointer	10	0.009981	[-0.0342, 0.0541]	0.9837
adaptive	pointer	100	0.01021	[-0.0339, 0.0544]	0.9815
adaptive	pointer	1000	0.01109	[-0.0331, 0.0552]	0.9716
adaptive	pointer	1e+04	0.01121	[-0.0329, 0.0554]	0.9700
adaptive	seq2seq	0.01	8.535×10^{-5}	[-0.124, 0.124]	1.0000
adaptive	seq2seq	0.1	0.003383	[-0.121, 0.127]	1.0000
adaptive	seq2seq	1	0.01157	[-0.113, 0.136]	0.9999
adaptive	seq2seq	10	-0.01745	[-0.142, 0.107]	0.9990

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	100	-0.1217	[-0.246, 0.00243]	0.0565	
adaptive	seq2seq	1000	-0.2032	[-0.327, -0.0791]	0.0005	*
adaptive	seq2seq	1e+04	-0.2032	[-0.327, -0.0791]	0.0005	*
tropical	pointer	0.01	0.002868	[-0.033, 0.0387]	1.0000	
tropical	pointer	0.1	0.01191	[-0.0239, 0.0477]	0.8963	
tropical	pointer	1	0.01148	[-0.0243, 0.0473]	0.9109	
tropical	pointer	10	0.008564	[-0.0273, 0.0444]	0.9780	
tropical	pointer	100	0.01037	[-0.0255, 0.0462]	0.9433	
tropical	pointer	1000	-0.002257	[-0.0381, 0.0336]	1.0000	
tropical	pointer	1e+04	-0.004148	[-0.04, 0.0317]	0.9997	
tropical	seq2seq	0.01	0.03828	[-0.102, 0.179]	0.9577	
tropical	seq2seq	0.1	0.03623	[-0.104, 0.177]	0.9678	
tropical	seq2seq	1	9.663×10^{-4}	[-0.14, 0.142]	1.0000	
tropical	seq2seq	10	-0.005852	[-0.146, 0.135]	1.0000	
tropical	seq2seq	100	-0.08697	[-0.228, 0.0537]	0.3852	
tropical	seq2seq	1000	-0.2035	[-0.344, -0.0629]	0.0022	*
tropical	seq2seq	1e+04	-0.2035	[-0.344, -0.0629]	0.0022	*
vanilla	pointer	0.01	0.001519	[-0.0543, 0.0573]	1.0000	
vanilla	pointer	0.1	0.001406	[-0.0544, 0.0572]	1.0000	
vanilla	pointer	1	0.001968	[-0.0538, 0.0577]	1.0000	
vanilla	pointer	10	0.002565	[-0.0532, 0.0583]	1.0000	
vanilla	pointer	100	0.003075	[-0.0527, 0.0588]	1.0000	
vanilla	pointer	1000	0.003735	[-0.052, 0.0595]	1.0000	
vanilla	pointer	1e+04	0.00502	[-0.0508, 0.0608]	0.9999	
vanilla	seq2seq	0.01	2.937×10^{-5}	[-0.0634, 0.0634]	1.0000	
vanilla	seq2seq	0.1	-0.001136	[-0.0645, 0.0623]	1.0000	
vanilla	seq2seq	1	0.007323	[-0.0561, 0.0707]	0.9997	
vanilla	seq2seq	10	0.001602	[-0.0618, 0.065]	1.0000	
vanilla	seq2seq	100	-0.13	[-0.193, -0.0666]	$< 10^{-4}$	*
vanilla	seq2seq	1000	-0.1588	[-0.222, -0.0954]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	-0.1586	[-0.222, -0.0952]	$< 10^{-4}$	*

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.475×10^{-5}	$[-3.13 \times 10^{-4}, 3.25 \times 10^{-4}]$	0.9936	0.00952	0.9538
adaptive	seq2seq	-0.04129	[-0.0614, -0.0229]	$< 10^{-4}$	-0.629	0.0002
tropical	pointer	-0.001803	[-0.00314, -5.81×10^{-4}]	0.2748	-0.143	0.2851
tropical	seq2seq	-0.04617	[-0.057, -0.0353]	$< 10^{-4}$	-0.632	0.0002
vanilla	pointer	5.811×10^{-4}	$[6.45 \times 10^{-5}, 0.00117]$	0.8150	0.0743	0.6007
vanilla	seq2seq	-0.03316	[-0.0393, -0.0273]	$< 10^{-4}$	-0.592	0.0002

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1006	[0.0562, 0.145]	0.0301	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.102	[0.059, 0.145]	0.0301	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1111	[0.0699, 0.152]	0.0256	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1101	[0.0638, 0.157]	0.0301	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1066	[0.066, 0.147]	0.0265	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1079	[0.0689, 0.147]	0.0247	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.09464	[0.0584, 0.131]	0.0265	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.09146	[0.0565, 0.126]	0.0265	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.02507	[-0.0985, 0.149]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.06333	[-0.0603, 0.187]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.06244	[-0.0586, 0.183]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01872	[-0.11, 0.148]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01762	[-0.097, 0.132]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.06808	[-0.0664, 0.203]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.01966	[-0.0849, 0.0455]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.01984	[-0.0848, 0.0451]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01332	[-0.00744, 0.0341]	0.5977	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.02208	[0.00321, 0.0409]	0.3140	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02271	[0.00628, 0.0391]	0.2403	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.02457	[0.00746, 0.0417]	0.2280	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.02074	[0.00451, 0.037]	0.2621	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02046	$[-5.88 \times 10^{-4}, 0.0415]$	0.3791	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.02068	[0.00187, 0.0395]	0.3360	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01952	[0.00488, 0.0341]	0.2494	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.02356	[-0.0381, -0.00899]	0.1638	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.0235	[-0.0322, -0.0148]	0.0269	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01904	[-0.0363, -0.00182]	0.3360	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01931	[-0.0426, 0.00394]	0.4121	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.04261	[-0.0894, 0.00421]	0.3894	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.01524	[-0.16, 0.13]	0.7850	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.06795	[-0.199, 0.0626]	0.6657	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.06813	[-0.199, 0.0628]	0.6657	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.08558	[0.0196, 0.152]	0.2276	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.09577	[0.049, 0.143]	0.0852	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.09299	[0.0391, 0.147]	0.1129	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.08722	[0.055, 0.119]	0.0353	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.113	[0.0505, 0.176]	0.1036	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.2175	[0.144, 0.291]	0.0270	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2998	[0.148, 0.451]	0.0869	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.3	[0.15, 0.45]	0.0869	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1243	[0.0283, 0.22]	0.2276	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.08884	[-0.0261, 0.204]	0.3177	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.09994	[-0.00806, 0.208]	0.3102	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1348	[0.0204, 0.249]	0.2276	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1387	[0.0418, 0.236]	0.1814	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.2216	[0.109, 0.334]	0.0869	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.3255	[0.285, 0.366]	0.0006	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.3236	[0.282, 0.365]	0.0006	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0487	[-0.0127, 0.11]	0.3177	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.05019	[-0.00938, 0.11]	0.3177	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.05124	[-0.00986, 0.112]	0.3177	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.04334	$[-2.04 \times 10^{-4}, 0.0869]$	0.3040	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.04966	[0.00879, 0.0905]	0.2276	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1818	[0.0763, 0.287]	0.1129	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.2112	[0.132, 0.291]	0.0353	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2123	[0.133, 0.291]	0.0353	*

A.7 id_exact

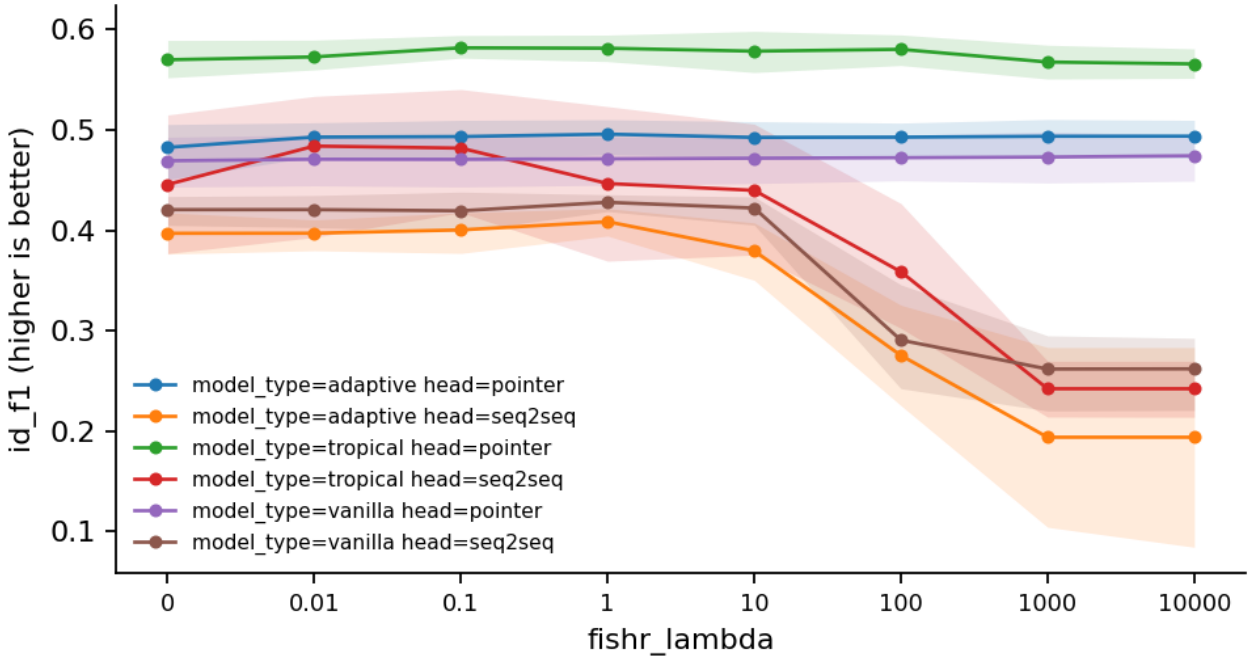


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

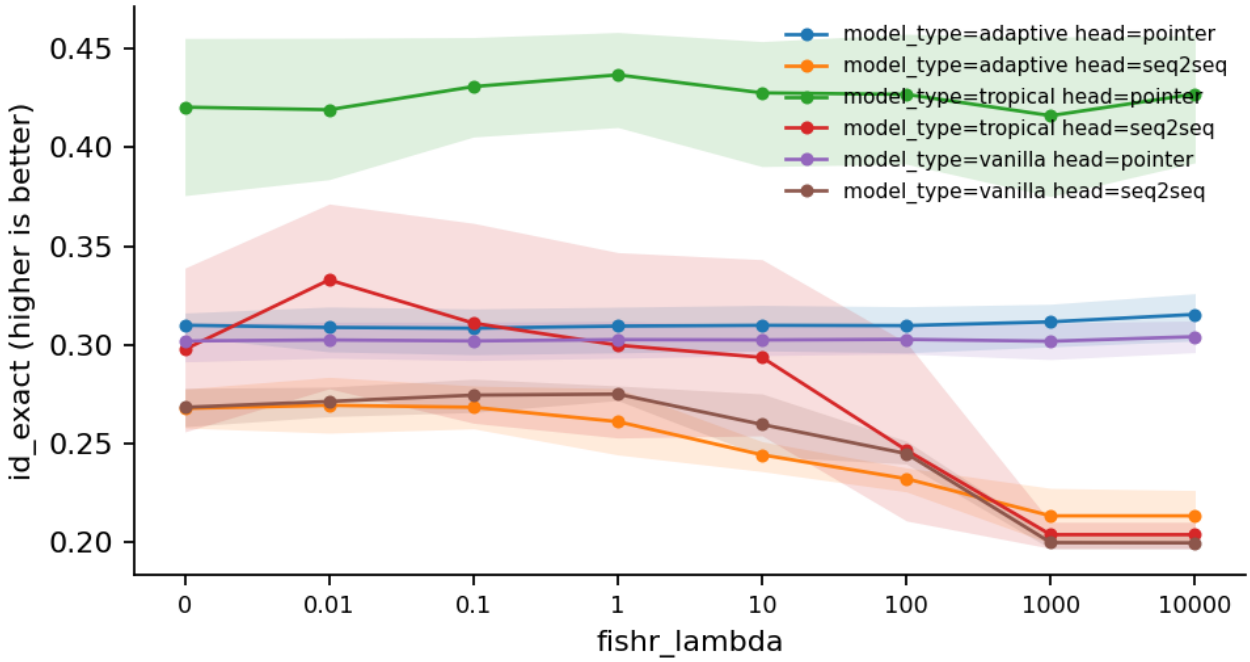


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.3099	[0.304, 0.316]	0.00807	5	0
adaptive	pointer	0.01	0.3088	[0.297, 0.319]	0.0149	5	0
adaptive	pointer	0.1	0.3084	[0.295, 0.318]	0.0153	5	0
adaptive	pointer	1	0.3095	[0.296, 0.319]	0.0155	5	0
adaptive	pointer	10	0.3099	[0.297, 0.32]	0.0155	5	0
adaptive	pointer	100	0.3098	[0.296, 0.319]	0.0154	5	0
adaptive	pointer	1000	0.3117	[0.299, 0.321]	0.0147	5	0
adaptive	pointer	1e+04	0.3154	[0.302, 0.326]	0.016	5	0
adaptive	seq2seq	0	0.2678	[0.258, 0.278]	0.0133	5	0
adaptive	seq2seq	0.01	0.2694	[0.255, 0.284]	0.0178	5	0
adaptive	seq2seq	0.1	0.2684	[0.258, 0.279]	0.0147	5	0
adaptive	seq2seq	1	0.2611	[0.244, 0.278]	0.0219	5	0
adaptive	seq2seq	10	0.2443	[0.236, 0.251]	0.00991	5	0
adaptive	seq2seq	100	0.2322	[0.226, 0.238]	0.00799	5	0
adaptive	seq2seq	1000	0.2134	[0.199, 0.227]	0.0178	5	0
adaptive	seq2seq	1e+04	0.2134	[0.199, 0.226]	0.0178	5	0
tropical	pointer	0	0.4204	[0.376, 0.455]	0.052	5	0
tropical	pointer	0.01	0.4191	[0.384, 0.455]	0.0465	5	0
tropical	pointer	0.1	0.4308	[0.405, 0.456]	0.0337	5	0
tropical	pointer	1	0.4368	[0.41, 0.458]	0.032	5	0
tropical	pointer	10	0.4276	[0.39, 0.454]	0.0415	5	0
tropical	pointer	100	0.4269	[0.392, 0.457]	0.0427	5	0
tropical	pointer	1000	0.4161	[0.375, 0.456]	0.0545	5	0
tropical	pointer	1e+04	0.4271	[0.392, 0.457]	0.0426	5	0
tropical	seq2seq	0	0.2977	[0.256, 0.339]	0.0557	5	0
tropical	seq2seq	0.01	0.3329	[0.278, 0.371]	0.0611	5	0
tropical	seq2seq	0.1	0.311	[0.26, 0.362]	0.0659	5	0
tropical	seq2seq	1	0.2999	[0.253, 0.347]	0.0615	5	0
tropical	seq2seq	10	0.2936	[0.254, 0.343]	0.0579	5	0
tropical	seq2seq	100	0.2466	[0.211, 0.301]	0.0618	5	0
tropical	seq2seq	1000	0.2038	[0.197, 0.21]	0.00849	5	0
tropical	seq2seq	1e+04	0.2038	[0.197, 0.21]	0.00849	5	0
vanilla	pointer	0	0.302	[0.291, 0.311]	0.013	5	0
vanilla	pointer	0.01	0.3025	[0.293, 0.312]	0.0122	5	0
vanilla	pointer	0.1	0.302	[0.292, 0.312]	0.0125	5	0
vanilla	pointer	1	0.3026	[0.293, 0.312]	0.0123	5	0
vanilla	pointer	10	0.3025	[0.295, 0.31]	0.0097	5	0
vanilla	pointer	100	0.3028	[0.295, 0.31]	0.00994	5	0
vanilla	pointer	1000	0.3018	[0.293, 0.311]	0.0121	5	0
vanilla	pointer	1e+04	0.3042	[0.296, 0.312]	0.0109	5	0
vanilla	seq2seq	0	0.2683	[0.259, 0.278]	0.0124	5	0
vanilla	seq2seq	0.01	0.2713	[0.264, 0.279]	0.0102	5	0
vanilla	seq2seq	0.1	0.2745	[0.266, 0.283]	0.0111	5	0
vanilla	seq2seq	1	0.2751	[0.272, 0.279]	0.00466	5	0
vanilla	seq2seq	10	0.2597	[0.244, 0.275]	0.0196	5	0
vanilla	seq2seq	100	0.2449	[0.24, 0.251]	0.00771	5	0
vanilla	seq2seq	1000	0.1998	[0.197, 0.203]	0.00435	5	0
vanilla	seq2seq	1e+04	0.1997	[0.197, 0.203]	0.00422	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.0011	[-0.0266, 0.0244]	1.0000	
adaptive	pointer	0.1	-0.00152	[-0.0271, 0.024]	1.0000	
adaptive	pointer	1	-4.400×10^{-4}	[-0.026, 0.0251]	1.0000	
adaptive	pointer	10	-4.000×10^{-5}	[-0.0256, 0.0255]	1.0000	
adaptive	pointer	100	-1.800×10^{-4}	[-0.0257, 0.0254]	1.0000	
adaptive	pointer	1000	0.00172	[-0.0238, 0.0273]	1.0000	
adaptive	pointer	1e+04	0.00548	[-0.0201, 0.031]	0.9877	
adaptive	seq2seq	0.01	0.00156	[-0.026, 0.0291]	1.0000	
adaptive	seq2seq	0.1	5.600×10^{-4}	[-0.027, 0.0281]	1.0000	
adaptive	seq2seq	1	-0.00672	[-0.0342, 0.0208]	0.9754	
adaptive	seq2seq	10	-0.02354	[-0.0511, 0.00397]	0.1186	
adaptive	seq2seq	100	-0.03564	[-0.0632, -0.00813]	0.0067	*
adaptive	seq2seq	1000	-0.05444	[-0.082, -0.0269]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	-0.05444	[-0.082, -0.0269]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.0013	[-0.0778, 0.0752]	1.0000	
tropical	pointer	0.1	0.01042	[-0.0661, 0.087]	0.9992	
tropical	pointer	1	0.01636	[-0.0602, 0.0929]	0.9880	
tropical	pointer	10	0.00724	[-0.0693, 0.0838]	0.9999	
tropical	pointer	100	0.00646	[-0.0701, 0.083]	1.0000	
tropical	pointer	1000	-0.0043	[-0.0808, 0.0722]	1.0000	
tropical	pointer	1e+04	0.00668	[-0.0699, 0.0832]	1.0000	
tropical	seq2seq	0.01	0.03518	[-0.057, 0.127]	0.8231	
tropical	seq2seq	0.1	0.01328	[-0.0789, 0.105]	0.9988	
tropical	seq2seq	1	0.00214	[-0.09, 0.0943]	1.0000	
tropical	seq2seq	10	-0.0041	[-0.0963, 0.0881]	1.0000	
tropical	seq2seq	100	-0.05108	[-0.143, 0.0411]	0.4972	
tropical	seq2seq	1000	-0.09388	[-0.186, -0.00172]	0.0444	*
tropical	seq2seq	1e+04	-0.09388	[-0.186, -0.00172]	0.0445	*
vanilla	pointer	0.01	4.800×10^{-4}	[-0.0198, 0.0208]	1.0000	
vanilla	pointer	0.1	0	[-0.0203, 0.0203]	1.0000	
vanilla	pointer	1	6.600×10^{-4}	[-0.0196, 0.021]	1.0000	
vanilla	pointer	10	5.000×10^{-4}	[-0.0198, 0.0208]	1.0000	
vanilla	pointer	100	8.000×10^{-4}	[-0.0195, 0.0211]	1.0000	
vanilla	pointer	1000	-1.600×10^{-4}	[-0.0205, 0.0201]	1.0000	
vanilla	pointer	1e+04	0.00218	[-0.0181, 0.0225]	0.9998	
vanilla	seq2seq	0.01	0.003	[-0.0154, 0.0214]	0.9975	
vanilla	seq2seq	0.1	0.0062	[-0.0122, 0.0246]	0.8891	
vanilla	seq2seq	1	0.00672	[-0.0116, 0.0251]	0.8481	
vanilla	seq2seq	10	-0.00868	[-0.027, 0.00967]	0.6542	
vanilla	seq2seq	100	-0.02344	[-0.0418, -0.00509]	0.0076	*
vanilla	seq2seq	1000	-0.06852	[-0.0869, -0.0502]	$< 10^{-4}$	*
vanilla	seq2seq	1e+04	-0.06866	[-0.087, -0.0503]	$< 10^{-4}$	*

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	9.457×10^{-4}	$[8.07 \times 10^{-4}, 0.00109]$	0.4354	0.19	0.1562
adaptive	seq2seq	-0.01096	[-0.016, -0.00587]	$< 10^{-4}$	-0.728	0.0002
tropical	pointer	-5.500×10^{-4}	$[-0.00223, 4.64 \times 10^{-4}]$	0.8708	0.0114	0.9368
tropical	seq2seq	-0.02338	[-0.0308, -0.0146]	$< 10^{-4}$	-0.615	0.0002
vanilla	pointer	1.757×10^{-4}	$[-4.41 \times 10^{-4}, 7.52 \times 10^{-4}]$	0.8448	0.0381	0.7908

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-0.01409	[-0.0158, -0.0124]	$< 10^{-4}$	-0.686	0.0002

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1184	[0.0672, 0.17]	0.0303	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.1166	[0.0733, 0.16]	0.0188	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1288	[0.0996, 0.158]	0.0038	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1341	[0.106, 0.162]	0.0030	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1252	[0.0844, 0.166]	0.0142	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1241	[0.0819, 0.166]	0.0147	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1143	[0.0611, 0.167]	0.0357	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1229	[0.0831, 0.163]	0.0142	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.02938	[-0.048, 0.107]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.06156	[-0.014, 0.137]	0.6919	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.03646	[-0.0556, 0.129]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.0248	[-0.0543, 0.104]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.03396	[-0.0289, 0.0968]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.00174	[-0.0758, 0.0793]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.00402	[-0.0106, 0.0186]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.00416	[-0.0104, 0.0188]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00796	[-0.00553, 0.0214]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00638	[-0.00642, 0.0192]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00644	[-0.00322, 0.0161]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00686	[-0.00574, 0.0195]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.00742	[-0.00395, 0.0188]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00698	[-0.00518, 0.0191]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.00984	$[7.68 \times 10^{-4}, 0.0189]$	0.6114	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.01126	$[-7.85 \times 10^{-4}, 0.0233]$	0.8447	

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-5.200×10^{-4}	[-0.00745, 0.00641]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00196	[-0.0141, 0.0102]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00616	[-0.016, 0.00365]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01396	[-0.0363, 0.00833]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01538	[-0.0447, 0.0139]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.01272	[-0.0243, -0.00112]	0.6114	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.01356	[-0.00667, 0.0338]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0137	[-0.00678, 0.0342]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.04212	[0.0217, 0.0625]	0.0621	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.03946	[0.0143, 0.0647]	0.0670	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.04004	[0.0124, 0.0676]	0.0670	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0484	[0.0216, 0.0752]	0.0670	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.06562	[0.0435, 0.0878]	0.0190	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.07758	[0.0586, 0.0966]	0.0069	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.09828	[0.0703, 0.126]	0.0112	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.102	[0.0743, 0.13]	0.0099	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1227	[0.0454, 0.2]	0.0670	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.0862	[0.036, 0.136]	0.0670	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1198	[0.0235, 0.216]	0.0670	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1369	[0.0712, 0.203]	0.0621	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.134	[0.0662, 0.202]	0.0621	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.1802	[0.091, 0.269]	0.0621	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2123	[0.149, 0.276]	0.0127	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.2232	[0.175, 0.271]	0.0043	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.03364	[0.0186, 0.0487]	0.0517	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.03112	[0.0125, 0.0497]	0.0670	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.02744	[0.00525, 0.0496]	0.0670	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.02758	[0.0118, 0.0433]	0.0670	

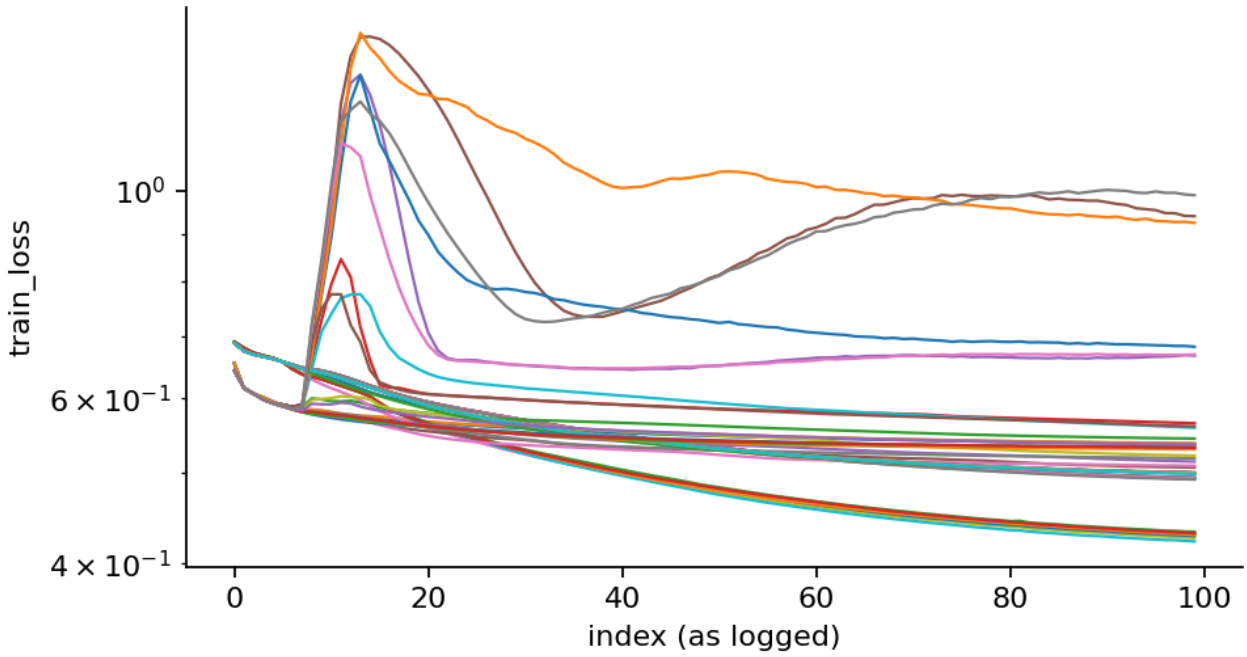


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.04282	[0.0218, 0.0638]	0.0621	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.05788	[0.0484, 0.0673]	0.0017	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.102	[0.0818, 0.122]	0.0033	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1045	[0.086, 0.123]	0.0022	*

B Curves

C Configuration

The base configuration every run shares (frozen hash `7237e0b98779`); factor levels are in Table 1.

```
task:
  name: MinCoinChangeDataset
  value_range:
  - 1
  - 10
  target_range:
  - 10
  - 20
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
```

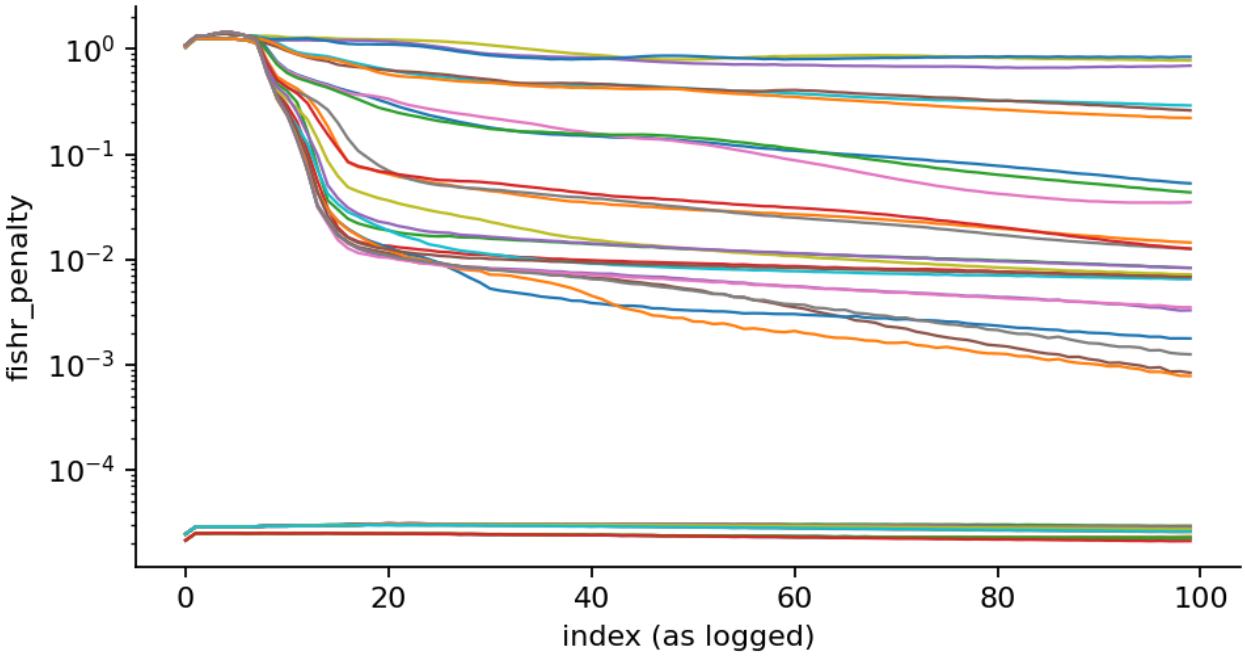


Figure 10: `fishr_penalty`, the mean over seeds per cell.

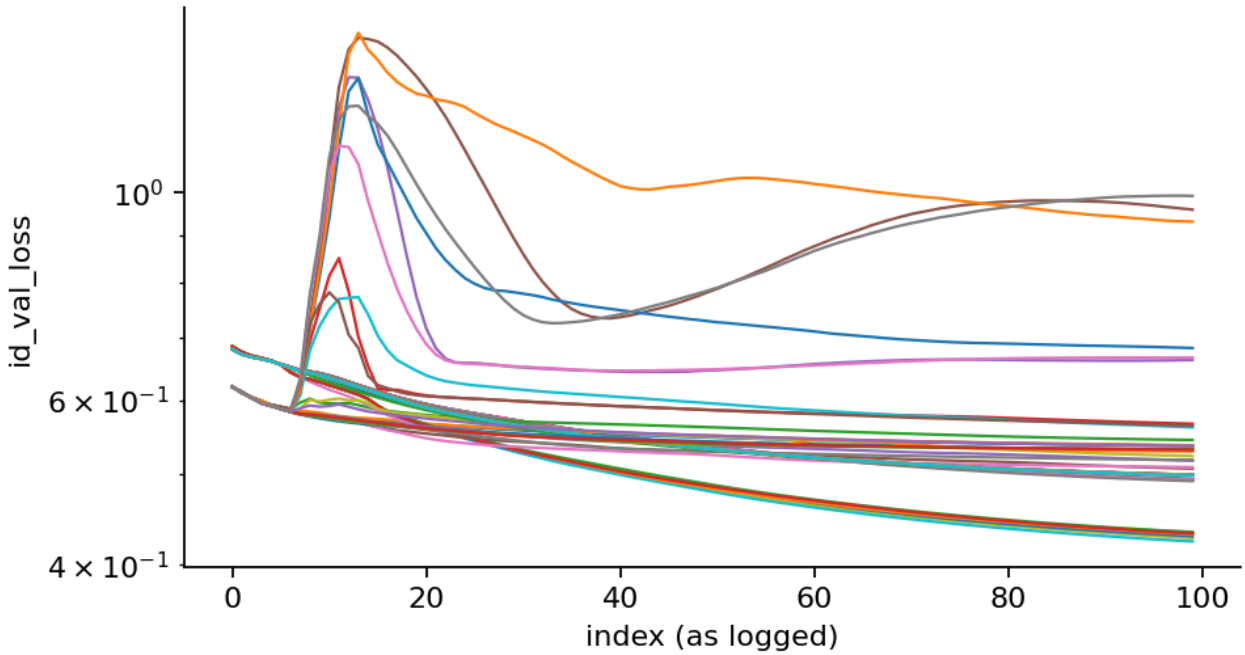


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/MinCoinChange/results`.